

# Native Response Geometry and Spectral Circulation in the Tri-Octagon Map

Finite-Ring SPD Self-Adjointness, Equivariant Spatial Observables,  
and State-Shell Attachment

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## Abstract

We study the exact linear response and geometric representations of an existing nonlinear complex-state map on cycles of length  $N = 3q$ . At a positive synchronized background, the complete amplitude and phase multipliers are real, even finite-ring Fourier symbols. Weak three-periodic inhomogeneity produces a positive weighted inner product for the pre-synchronization phase response, with weights equal to squared background amplitudes. Composition with the uniform phase synchronizer generically violates every positive diagonal balance condition. This local obstruction has different spectral consequences in different finite sectors. The positive triad has exactly real transverse spectrum and, near a strictly stable equal background, admits a generally non-diagonal positive-definite symmetrizer. For  $N \geq 9$ , three-distinct repeated backgrounds satisfying an explicit nondegeneracy condition force non-real Bloch modes. Separately, a six-ring admits an exact rank-one square-zero response and a real/defective/complex unfolding using only an existing parameter. We then classify low-degree spatial equivariants and finite shell-carrier assignments, showing that symmetry does not select a unique position observable. The adopted face decoder is an invertible representation on three tangent fibres, while intensities and chirality generally fail autonomous closure. Full complex coherence and the real Gram matrix do close on the nonzero-pre-synchronization domain, with an exact obstruction at the native zero convention. These are finite-dimensional mathematical results; no physical metric, current, motion or boundary dynamics is identified.

**Scope.** Synthesis of the completed GR0, GR1, GR2, CM0 and SA0 checkpoints. No scientific source changes or new physical model. The retained evidence comprises 429 heterogeneous exact/source checks, supplemented by separately reported numerical evidence. It is not a count of theorems.

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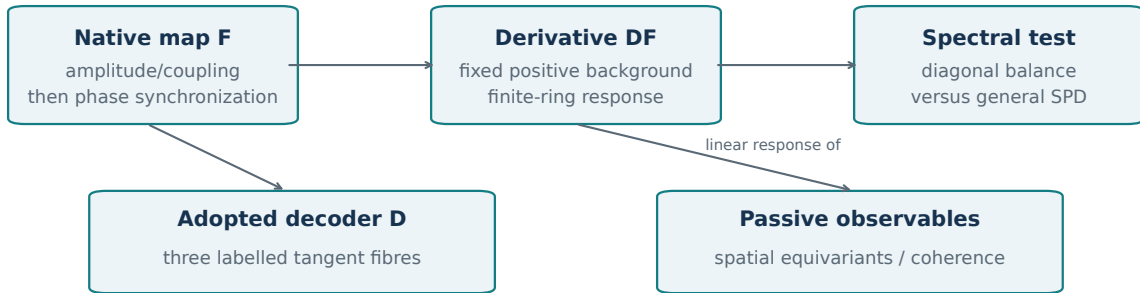
# 1 Introduction

The Tri-Octagon programme contains several different kinds of geometry: a fixed folded shell, a complex state represented in selected face tangent planes, and graph operators governing a nonlinear recurrence. A comparison between these structures is useful only after their domains and transformations have been stated. In particular, a background-dependent inner product for a response operator is not automatically a metric on the shell, and a vector readout is not automatically the location of an object.

This paper consolidates five completed investigations [1, 2, 3, 4, 5]. The sequence begins with the exact homogeneous response, proceeds to inhomogeneous weighted response and its reversibility obstruction, resolves the finite-sector spectral alternatives, and ends with the multiplicity and closure of spatial representations. The principal conclusion is conditional: a local obstruction to diagonal reversibility can coexist with self-adjointness in a non-diagonal inner product, but other native finite-ring responses have spectral obstructions to every positive-definite symmetrizer.

The recurrence and its positive fixed equations predate these investigations [6]; the adopted tangent-state correspondence and matched transport are also prior results [7, 8]. The equal-unit triad derivative appears in Paper F [11]. These are foundations, not new claims of this synthesis. The ledger in Appendix B and the accompanying source/proof crosswalk distinguish inherited definitions, checkpoint deductions, numerical evidence, interpretation limits and open questions. No claim of historical priority is made.

## Native map, response, and representation



Derivation: GR0-GR2, CM0, SA0. No arrow denotes a physical interaction.

Figure 1: Logical architecture. Solid arrows are native updates or exact mathematical deductions; the separate representation/readout branch does not feed geometry into the recurrence. Source: GR0-GR2 and SA0; definitions and proofs in Sections 2–13. No arrow denotes a physical coupling.

## 2 Native map and mathematical setting

For  $N = 3q \geq 3$ , indices are cyclic and

$$(\Delta_N z)_i = z_{i-1} + z_{i+1} - 2z_i. \quad (2.1)$$

For  $N = 3$  the two neighbours are the other two channels. Native coefficients are real and repeat with period three. The complete map  $F$  is

$$\tilde{z}_i = z_i + \varepsilon z_i(k_i - |z_i|^2) + g(\Delta_N z)_i, \quad (2.2)$$

$$\tilde{\phi}_i = \text{Arg}_0(\tilde{z}_i), \quad \delta_i = \lambda \sum_{j \sim i} \sin 3(\tilde{\phi}_j - \tilde{\phi}_i), \quad (2.3)$$

$$F_i(z) = |\tilde{z}_i| \exp(i(\tilde{\phi}_i + \delta_i)). \quad (2.4)$$

Here  $\text{Arg}_0(0) = 0$ ; when  $\lambda = 0$  synchronization is the identity. Set  $\ell = 3\lambda$  as an abbreviation, not a time increment. All derivatives below are taken where the relevant pre-synchronization entries are nonzero. Unit complex factors express the sine terms analytically on this domain, without a principal-argument cut. No differentiability at zero is silently assumed.

In the triad, the pre-stage is  $z - \nabla_{\mathbb{R}^6} \mathcal{V}$ , where

$$\mathcal{V} = \varepsilon \sum_i \left( \frac{|z_i|^4}{4} - \frac{k_i |z_i|^2}{2} \right) + \frac{g}{2} \sum_{i < j} |z_i - z_j|^2. \quad (2.5)$$

Indeed differentiation with respect to each real or imaginary component gives the negative of the corresponding increment in (2.2). If that increment is  $b$ , synchronization preserves moduli and therefore

$$I^+ - I = 2\varepsilon \sum_i (k_i |z_i|^2 - |z_i|^4) - 2g \sum_{i < j} |z_i - z_j|^2 + \|b\|^2, \quad I = \sum_i |z_i|^2. \quad (2.6)$$

This is an exact budget, not a conservation or descent claim: at  $\varepsilon = 1, g = 0, k_i = 1$ , the common real input 2 maps to  $-4$ , taking  $I$  from 12 to 48 and  $\mathcal{V}$  from 6 to 168. No continuous gradient flow or physical energy is substituted for the native discrete map. *Checkpoint authority: GR0 §4; prior potential normalization in Papers D/E and Atlas 07.*

### 3 Homogeneous finite-ring response

**Theorem 3.1** (Fixed circle and complete derivative). *Let  $k_i = k > 0, \varepsilon \neq 0$ . The nonzero fixed states on the common-channel line form the circle  $z_* = \sqrt{k}e^{i\chi}\mathbf{1}$ . In the local chart  $z = e^{i\chi}(\sqrt{k}\mathbf{1} + a + ib)$ , the real derivative has blocks*

$$A_0 = (1 - 2\varepsilon k)I + g\Delta_N, \quad J_{\phi,0} = (I + \ell\Delta_N)(I + g\Delta_N). \quad (3.1)$$

*The amplitude perturbation is  $a$ ; the phase perturbation is  $b/\sqrt{k}$ .*

*Proof.* On the common line, graph coupling vanishes and synchronization does nothing. A nonzero fixed value  $w$  satisfies  $\varepsilon(k - |w|^2) = 0$ . At the positive radius,  $|\sqrt{k} + a_i + ib_i|^2 = k + 2\sqrt{k}a_i + O(\|(a, b)\|^2)$ . Differentiating the cubic term gives a real increment  $-2\varepsilon k a_i$  and zero imaginary increment. The pre-blocks are therefore  $A_0$  and  $I + g\Delta_N$ . Differentiating  $\sin 3(\phi_j - \phi_i)$  at equal phase gives  $\ell\Delta_N\phi$ . The synchronizer changes no modulus, so its derivative leaves the radial block unchanged and composes after the phase pre-block.  $\square$

For  $\varepsilon = 0$  every common nonzero amplitude is fixed, giving another neutral direction. For  $\varepsilon \neq 0, k \leq 0$  the stated nonzero fixed family does not exist.

**Theorem 3.2** (Exact finite spectrum and stability). *With  $h_m = 4\sin^2(\pi m/N)$ ,  $m = 0, \dots, N-1$ ,*

$$\boxed{\mu_a(m) = 1 - 2\varepsilon k - gh_m, \quad \mu_\phi(m) = (1 - gh_m)(1 - \ell h_m).} \quad (3.2)$$

*Strict normal linear contraction is equivalent to*

$$|\mu_a(m)| < 1 \quad \text{all } m, \quad |\mu_\phi(m)| < 1 \quad m \neq 0. \quad (3.3)$$

*For each fixed finite  $N$  these inequalities also give local attraction to some point on the fixed circle.*

*Proof.* The Fourier vector  $e^{2\pi i m i/N}$  has Laplacian eigenvalue  $-h_m$ . Substitution into the two polynomial blocks proves (3.2). The blocks are real symmetric, so on the normal space the Euclidean norm is the largest multiplier modulus. The excluded phase mode has multiplier one. In a local common-phase quotient chart the derivative norm is strictly less than one; continuity gives a smaller contracting neighbourhood. Phase drifts are smooth, vanish on the fixed circle and are bounded by the geometrically decaying transverse deviations, hence summable. The original orbit converges to a nearby phase on the circle.  $\square$

There is no claim of uniform basin size as  $N$  grows. Unit-modulus boundary cases require further analysis. The exact amplitude criterion is

$$0 < \varepsilon k < 1, \quad 0 < 2\varepsilon k + g h_{\max} < 2,$$

where  $h_{\max} = 4$  for even  $N$  and  $4 \cos^2(\pi/(2N))$  for odd  $N$ . It follows by checking the attained endpoints of an affine symbol. The phase criterion is the finite list

$$0 < (g + \ell)h_m - g\ell h_m^2 < 2, \quad m \neq 0.$$

These statements permit either sign of the native coefficients; a sufficient all-finite-ring domain is  $0 \leq g < 1/2$ ,  $0 \leq \lambda < 1/6$ ,  $g + 3\lambda > 0$ ,  $0 < \varepsilon k < 1 - 2g$ .

The stable response is relaxational in the precise spectral sense: its normal multipliers are real with modulus below one, possibly alternating in sign. They are even in wave angle and have no propagating conjugate branch  $e^{\pm i c \theta}$ . This excludes such a linear branch at this homogeneous background, not every nonlinear travelling state.

### 3.1 Formal symbol and restricted modal limit

Write  $D_\phi = g + \ell$ . The analytic symbol obeys

$$h(\theta) = \theta^2 - \theta^4/12 + O(\theta^6), \quad \mu_\phi = 1 - D_\phi \theta^2 + (D_\phi/12 + g\ell)\theta^4 + O(\theta^6).$$

At  $D_\phi = 0$ ,  $\ell = -g$  and  $\mu_\phi = 1 - g^2 h^2$ , with quartic rather than quadratic leading decrement when  $g \neq 0$ .

For  $D_\phi > 0$ , a fixed Fourier index  $m$  and bounded  $0 \leq \tau \leq T$ , the exact controlled statement is

$$\mu_\phi(2\pi m/N)^{\lfloor \tau N^2 \rfloor} = e^{-4\pi^2 D_\phi m^2 \tau} + O_{m,T}(N^{-2}). \quad (3.4)$$

To prove it, expand  $\log \mu_\phi = -4\pi^2 D_\phi m^2 N^{-2} + O_m(N^{-4})$ ; multiplying by the integer count leaves a uniformly  $O(N^{-2})$  exponent error. The  $m=0$  identity is exact. This controls fixed finitely many modes of the linearization, not the complete nonlinear map, all growing mode indices or a physical continuum limit. The triad pullback occupies  $h = 0, 3, 3$ , not a long-wave sector. *Checkpoint authority: GR0 §§5–6, L05–L09.*

## 4 Weak unequal-background branch

Let  $k_i = k + \eta\kappa_i$ ,  $k > 0$ ,  $\kappa_{i+3} = \kappa_i$  and  $\sum_{i=1}^3 \kappa_i = 0$ . Assume (3.3) for the chosen finite  $N$  at  $\eta = 0$ .

**Theorem 4.1** (Native branch and first response). *There is a unique nearby positive real-analytic in-phase radius branch, modulo common phase, satisfying*

$$\varepsilon r_i(k_i - r_i^2) + g(\Delta_N r)_i = 0. \quad (4.1)$$

*It is three-periodic and  $r = \sqrt{k}\mathbf{1} + \eta u + O(\eta^2)$ , with*

$$(2\varepsilon k I - g\Delta_N)u = \varepsilon\sqrt{k}\kappa, \quad u_i = \frac{\varepsilon\sqrt{k}}{2\varepsilon k + 3g}\kappa_i. \quad (4.2)$$

*Proof.* The derivative in  $r$  of (4.1) at the equal state is  $-2\varepsilon k I + g\Delta_N$ . Its negative has eigenvalues  $1 - \mu_a(m) \in (0, 2)$ , hence is invertible. The analytic implicit-function theorem and positivity by continuity give the branch. Differentiation in  $\eta$  gives the first equation. Uniqueness and shift-by-three symmetry force three-periodicity. On the repeated zero-sum triad sector,  $\Delta_N \kappa = -3\kappa$ , giving the explicit solution. Strict noncommon stability persists locally for this fixed finite  $N$ .  $\square$

Distinct  $\kappa_i$  give distinct radii for small nonzero  $\eta$ . Two equal coefficients give equal radii by uniqueness of the reduced triad branch; the converse follows by subtracting its positive fixed equations. The zero-mean assertion applies to  $u$ , not all higher corrections.

For  $g > 0$ , (4.2) has the mathematical screened-Poisson or modified-Helmholtz form  $2\varepsilon k I + g(-\Delta_N)$ . This is a structural analogy only:  $\kappa$  is not an established mass density,  $u$  is not a gravitational potential, and graph index is not established physical space. For negative  $g$  the finite-system inverse still exists under the stated hypothesis, but the usual positive graph-stiffness interpretation is unavailable. *Checkpoint authority: GR1 §2, (2)–(6), L10.*

## 5 Weighted response geometry and stage order

At any positive fixed branch let

$$R = \text{diag}(r_i), \quad H_r = I + \varepsilon \text{diag}(k_i - r_i^2) + g\Delta_N, \quad H_r r = r. \quad (5.1)$$

The real amplitude and imaginary pre-blocks are  $H_r - 2\varepsilon R^2$  and  $H_r$ , by direct differentiation of the native cubic. In phase coordinates  $b = R\phi$ ,

$$P = R^{-1}H_r R, \quad P_{ij} = g r_j / r_i \ (j \sim i), \quad P_{ii} = 1 - g \sum_{j \sim i} r_j / r_i, \quad P\mathbf{1} = \mathbf{1}. \quad (5.2)$$

**Theorem 5.1** (Native weighted self-adjointness). *On every positive branch,*

$$R^2 P = P^T R^2, \quad \langle f, h \rangle_r = \sum_i r_i^2 f_i h_i, \quad \langle f, (I - P)f \rangle_r = g \sum_{\{i,j\}} r_i r_j (f_i - f_j)^2. \quad (5.3)$$

*For  $g \neq 0$ , the diagonal weights are unique up to scale on the connected graph.*

*Proof.*  $R^2 P = R H_r R$  is symmetric. Edgewise,  $r_i^2 P_{ij} = g r_i r_j = r_j^2 P_{ji}$ . Any other diagonal balancing weights must have  $w_i / r_i^2 = w_j / r_j^2$  along every edge, proving uniqueness by connectedness. Using the row sum and pairing the two orientations of each edge gives the quadratic identity.  $\square$

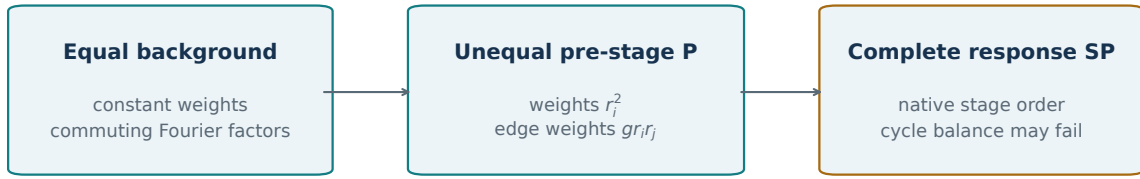
For  $g > 0$  these are positive edge conductances; for  $g < 0$  balance survives but the form has the opposite sign. At  $g=0$ ,  $P = I$  and any positive diagonal weight works. Row sums do not by themselves imply a Markov matrix.

The native synchronizer differentiates to  $S_\phi = I + \ell \Delta_N$ . Therefore

$$\boxed{J_\phi = S_\phi P, \quad DF|_{(a,\phi)} = \text{diag}(H_r - 2\varepsilon R^2, S_\phi P), \quad J_\phi \mathbf{1} = \mathbf{1}.} \quad (5.4)$$

In Cartesian imaginary coordinates the second block is instead  $RS_\phi R^{-1}H_r$ . The native order is  $S_\phi P$ , not  $PS_\phi$ . Both factors have self-adjoint descriptions, but usually in different inner products. Their product requires an independent reversibility test.

### Weighted response before and after synchronization



$R^2 P = P^T R^2$  does not imply a positive diagonal balance for  $SP$ .

Derivation: GR1 §§3-6. Conductance language assumes  $g > 0$ ; no spacetime identification.

Figure 2: Background-dependent weighted response and the stage-order obstruction. Edge labels  $gr_i r_j$  are the exact conductances of (5.3) for  $g$  positive. Composition is the native order (5.4). The final line records the theorem in Section 6, not a probability-current or curvature interpretation. Source: GR1 §§3-6.

## 6 Synchronization and diagonal cycle obstruction

For the triad set  $r = (a, b, c) > 0$ ,  $\sigma_1 = a + b + c$ ,  $\sigma_2 = ab + ac + bc$ ,  $\sigma_3 = abc$ ,  $d = 1 - 3\ell$ . Since  $S_\phi = dI + \ell \mathbf{1}\mathbf{1}^T$ ,

$$J_{ij} = dg r_j / r_i + \ell v_j \quad (i \neq j), \quad v_j = \sum_i P_{ij} = 1 - g\sigma_1 / r_j + gr_j \sum_i 1 / r_i. \quad (6.1)$$

**Theorem 6.1** (Triad diagonal criterion). *In a sufficiently small neighbourhood of any strictly stable equal triad, positive diagonal  $W$  satisfying  $WJ = J^T W$  exists exactly when*

$$\mathcal{C}_{\text{cyc}} = J_{12}J_{23}J_{31} - J_{21}J_{32}J_{13} \quad (6.2)$$

$$= \frac{g\ell d(a-b)(a-c)(b-c)}{a^2 b^2 c^2} \mathcal{K} = 0, \quad (6.3)$$

$$\mathcal{K} = g^2 \ell \sigma_1^3 + [g^2(1 - 2\ell) - g\ell]\sigma_1 \sigma_2 + (3g\ell - g + \ell)\sigma_3. \quad (6.4)$$

*Proof.* At the equal background all off-diagonals equal  $q_0 = g + \ell - 3g\ell$ . Its transverse multiplier is  $1 - 3q_0$ , so strict stability implies  $0 < q_0 < 2/3$ . Off-diagonals remain positive locally. Multiplying pairwise balance around the triangle proves necessity. If the cycle difference is zero, the positive choice

$$W = \text{diag}(J_{21}J_{31}, J_{12}J_{31}, J_{13}J_{21}) \quad (6.5)$$

balances edges 12 and 13 identically; the edge-23 residual is precisely  $\mathcal{C}_{\text{cyc}}$ .

For the factorization substitute the displayed J entries. The cleared numerator is alternating in a,b,c, hence divisible by their Vandermonde product. Single-stage terms and the d=0 terms cancel. After extracting  $g\ell d$  and the denominator, the remaining symmetric cubic has basis  $\sigma_1^3, \sigma_1\sigma_2, \sigma_3$ . Collecting coefficients gives respectively  $g^2\ell$ ,  $g^2(1-2\ell) - g\ell$ , and  $3g\ell - g + \ell$ . This proves the rational identity as well as the local criterion.  $\square$

Thus the exceptional local loci are  $g = 0$ ,  $\ell = 0$ ,  $d = 0$ , equal pairs of radii, and  $\mathcal{K} = 0$ . The first two are globally reversible on positive branches with weights I and  $R^2$ , respectively; equal radii give a symmetric J. At  $\ell = 1/3$ ,  $J = \mathbf{1}v^T/3$ , balanced by  $\text{diag}(v)$  when  $v_i > 0$ , particularly locally. For two equal radii  $(a, a, c)$ , writing  $J_{13} = J_{23} = B$ ,  $J_{31} = J_{32} = C$ , the local weights can be taken as  $\text{diag}(1, 1, B/C)$ .

The additional locus is exact, not fitted. Write

$$\mathcal{K} = A_0 + \ell B_0, \quad A_0 = g(g\sigma_1\sigma_2 - \sigma_3), \quad B_0 = g^2(\sigma_1^3 - 2\sigma_1\sigma_2) - g\sigma_1\sigma_2 + (3g + 1)\sigma_3.$$

If  $B_0 \neq 0$  its location is  $-A_0/B_0$ . At radii  $(999/1000, 1, 1001/1000)$  and  $g=1/10$  it is  $\ell = 333331/16333302$ , with positive off-diagonals. There is no omitted  $A_0 = B_0 = 0$  positive-radius case: for  $g \neq 0$ ,  $A_0 = 0$  gives  $g = 1/p$ ,  $p = \sigma_1\sigma_2/\sigma_3 > 0$ ; writing  $q = \sigma_1^3/\sigma_3$ , one obtains  $B_0/\sigma_3 = (q + p)/p^2 > 0$ . For  $g=0$ ,  $B_0 = \sigma_3 > 0$ .

### 6.1 Weak onset and the stronger full-ring restriction

Set  $K_0 = g(9g - 1) + \ell(1 - 3g)^2$ , and  $V(\kappa) = \prod_{i < j} (\kappa_i - \kappa_j)$ . Along Theorem 4.1,

$$\mathcal{C}_{\text{cyc}} = g\ell d K_0 \left( \frac{\varepsilon}{2\varepsilon k + 3g} \right)^3 V(\kappa) \eta^3 + O(\eta^4). \quad (6.6)$$

This follows by substituting the branch derivative into the three difference factors and  $\mathcal{K} = k^{3/2} K_0$  at equality. At  $K_0 = 0$ , symmetry and  $\sum u_i = 0$  force the first variation of  $\mathcal{K}$  to vanish, leaving  $O(\eta^5)$  or smaller. A first-order fitted weight cannot settle the exact triad obstruction.

**Theorem 6.2** (Full-ring diagonal restriction). *For  $N = 3q \geq 6$ , positive repeated radii and  $g\ell \neq 0$ , three distinct radii forbid every positive diagonal symmetrizer. For  $(a, a, c)$ , one exists exactly when*

$$\ell - g - g\ell(2a/c + c/a) = 0, \quad (6.7)$$

*in which case  $W=R$  works. Equal radii,  $g=0$  or  $\ell = 0$  retain their preceding reversible cases.*

*Proof.* Average any putative positive diagonal W over shift by three; it remains positive and balancing, and becomes three-periodic. Distance-two entries have one intermediate node:

$$J_{i,i+2} = \ell g r_{i+2}/r_{i+1}, \quad J_{i+2,i} = \ell g r_i/r_{i+1}.$$

They force  $w_i/r_i = w_{i+2}/r_{i+2}$  through all three residue classes, so the averaged weight is proportional to R. The remaining distance-one residual is

$$(RJ - J^T R)_{i,i+1} = (a - b)[\ell - g + 3g\ell - g\ell\sigma_1(1/a + 1/b)]. \quad (6.8)$$

For three distinct radii all three brackets must vanish; subtracting two gives  $-g\ell\sigma_1(1/a - 1/c) \neq 0$ , a contradiction. If  $a=b$ , the remaining brackets reduce to (6.7). All other entries already balance or vanish, proving sufficiency too.  $\square$

No positivity of J entries or stability hypothesis is needed for this ring theorem. The triad  $d = 0$  and accidental  $\mathcal{K} = 0$  loci therefore do not generally rescue the full ring. *Checkpoint authority: GR1 §§4–6, L13–L16.*

## 7 General SPD self-adjointness

Because “symmetrizable” often denotes positive diagonal symmetrization in matrix literature, we use *diagonal balance* for that stronger node-local property and *SPD self-adjointness* for  $GJ = J^T G$  with arbitrary  $G > 0$ . See, for example, the positive-diagonal convention in [19, Definition 4.1].

**Theorem 7.1** (Finite real criterion). *A real square  $J$  admits a real symmetric positive-definite  $G$  with  $GJ = J^T G$  if and only if  $J$  is diagonalizable over  $\mathbb{R}$  with entirely real spectrum.*

*Proof.* If  $G$  exists,  $Q = G^{1/2}$  makes  $QJQ^{-1}$  real symmetric, since  $(QJQ^{-1})^T = Q^{-1}J^T Q = QJQ^{-1}$ . Conversely, write  $J = V\Lambda V^{-1}$  with  $V$  real invertible and  $\Lambda$  real diagonal. For any SPD  $K$  commuting with  $\Lambda$ ,

$$G = V^{-T} K V^{-1} \quad (7.1)$$

is positive and balances  $J$ . These are all solutions: multiplying balance by  $V^T$ ,  $V$  gives  $K\Lambda = \Lambda K$ .  $\square$

Repeated semisimple roots and zero eigenvalues are allowed. An alternative, avoiding basis choices inside eigenspaces, is

$$G = \sum_{\rho} c_{\rho} E_{\rho}^T E_{\rho}, \quad E_{\rho} = \prod_{\sigma \neq \rho} \frac{J - \sigma I}{\rho - \sigma}, \quad c_{\rho} > 0. \quad (7.2)$$

Here  $\sum E_{\rho} = I$  proves positivity, and  $E_{\rho} J = \rho E_{\rho}$  proves balance. In general these forms mix nodes and are nonunique. Positive diagonal detailed balance is strictly stronger than arbitrary SPD self-adjointness. Neither existence nor nonuniqueness selects a physical geometry. *Checkpoint authority: GR2 §2, L17; elementary lemma, no priority claim.*

## 8 Exact positive-triad spectrum

Define

$$\sigma = a + b + c, \quad h = \sigma(1/a + 1/b + 1/c) - 3, \quad t = \frac{\sigma(a^2 + b^2 + c^2)}{abc}, \quad d = 1 - 3\ell. \quad (8.1)$$

The invariant  $t$  is not time. The symmetric matrix

$$M_3 = \sigma R^{-1} - \mathbf{1}\mathbf{1}^T, \quad H_r = I - gM_3 \quad (8.2)$$

has kernel  $r$ . Weighted Cauchy–Schwarz gives

$$v^T M_3 v = \sigma \sum_i v_i^2 / r_i - (\sum_i v_i)^2 \geq 0,$$

with equality precisely for  $v \parallel r$ . Its positive eigenvalues  $\mu_{\pm}$  therefore have sum  $h$  and product  $t$ , by trace and principal minors.

**Theorem 8.1** (Reality of the transverse quotient). *The characteristic polynomial is*

$$\det(\xi I - J) = (\xi - 1)Q(\xi), \quad (8.3)$$

$$Q(\xi) = \xi^2 - d(2 - gh)\xi + d^2(1 - gh + g^2 t), \quad (8.4)$$

$$\mathcal{D}_3 = \boxed{d^2 g^2 (h^2 - 4t) \geq 0}, \quad \rho_{\pm} = d(1 - g\mu_{\pm}). \quad (8.5)$$

*The transverse quotient is always real semisimple. Its roots are distinct exactly when  $gd \neq 0$  and the positive radii are not all equal.*

*Proof.* The transverse object is  $\mathbb{R}^3/\text{span}\{\mathbf{1}\}$ ; the Euclidean sum-zero plane need not be J-invariant. P is similar to symmetric  $I - gM_3$ ; its noncommon eigenvectors span this quotient. The synchronizer  $dI + \ell\mathbf{1}\mathbf{1}^T$  acts as scalar d on it, proving the polynomial and semisimplicity. The two positive eigenvalues of  $M_3$  coincide only at equal radii. If both equal  $\mu$ , then  $M_3 = \mu(I - rr^T/(r^T r))$ . Its off-diagonal entries are all  $-1$ , forcing  $ab=ac=bc$  and hence equality. The converse follows by substitution. Cases  $g=0$  and  $d=0$  yield scalar quotient action.  $\square$

### 8.1 Full-matrix defects and local SPD geometry

Let  $w_\alpha \perp r$  be eigenvectors of  $M_3$ ,  $v_\alpha = R^{-1}w_\alpha$ , and  $p_\alpha = 1 - g\mu_\alpha$ . In the basis  $(\mathbf{1}, v_-, v_+)$ ,

$$J \sim \begin{pmatrix} 1 & \ell p_- \mathbf{1}^T v_- & \ell p_+ \mathbf{1}^T v_+ \\ 0 & dp_- & 0 \\ 0 & 0 & dp_+ \end{pmatrix}. \quad (8.6)$$

This follows by applying P and then  $dI + \ell\mathbf{1}\mathbf{1}^T$  to each vector. Thus, for nonconstant radii and  $g\ell d \neq 0$ , a defect occurs exactly when

$$dp_\alpha = 1, \quad \mathbf{1}^T v_\alpha \neq 0. \quad (8.7)$$

It is a size-two Jordan block at one, not a defective transverse coincidence.

To classify the vanishing coupling, multiply  $M_3 w = \mu w$  by  $\mathbf{1}^T$ :

$$\mathbf{1}^T R^{-1} w = (\mu + 3) \mathbf{1}^T w / \sigma.$$

If this is zero, the eigen-equation reduces to  $\sigma R^{-1} w = \mu w$ . A nonzero sum-zero  $w$  can then be supported only on equal radii. Consequently every resonance for three distinct radii is defective; for exactly two equal radii the antisymmetric equal-pair mode is semisimple at resonance and the other mode is defective. Equal radii and cases  $g = 0, \ell = 0, d = 0$  are diagonalizable; at  $d=0$ ,  $J^2 = J$ .

Near a strictly stable equal background the transverse eigenvalues remain inside the unit disk and away from one. These resonances are excluded, so Theorem 7.1 supplies an SPD metric throughout a sufficiently small positive unequal triad branch, even when Theorem 6.1 excludes every positive diagonal weight.

Finally, let  $v_i = u_i/\sqrt{k}$  in the weak expansion. The first variation of  $M_3$ , restricted to the equal-background transverse plane, is  $-3\Pi \text{diag}(v)\Pi$ ,  $\Pi = I - \mathbf{1}\mathbf{1}^T/3$ . It is traceless and has squared eigenvalue gap  $6 \sum v_i^2$ . Therefore

$$\mathcal{D}_3 = 6d^2 g^2 \left( \frac{\varepsilon}{2\varepsilon k + 3g} \right)^2 \sum_i \kappa_i^2 \eta^2 + O(\eta^3). \quad (8.8)$$

The positive quadratic onset splits real roots; no order makes the positive-triad transverse spectrum complex. This is independent of the cubic cycle obstruction. *Checkpoint authority: GR2 §§3–4, L18–L20.*

## 9 Exact finite Bloch decomposition

Number nodes by cell  $n = 0, \dots, q-1$  and residue  $\alpha = 0, 1, 2$ . For  $\phi_{n,\alpha} = z^n v_\alpha$ ,  $z = e^{i\theta}$ ,  $\theta = 2\pi m/q$ ,

$$D(z) = \begin{pmatrix} -2 & 1 & z^{-1} \\ 1 & -2 & 1 \\ z & 1 & -2 \end{pmatrix}, \quad (9.1)$$

$$H_r(z) = I - g \operatorname{diag}((\sigma - 3r_\alpha)/r_\alpha) + gD(z), \quad (9.2)$$

$$J(z) = \boxed{(I + \ell D(z))R^{-1}H_r(z)R}. \quad (9.3)$$

Substitution at cell boundaries supplies the  $z^{\pm 1}$  factors. Fourier transformation in  $n$  is an exact finite direct sum. Both  $D$  and  $H_r$  are Hermitian on the unit circle, but the product in (9.3) need not be. Conjugate phases give conjugate blocks.

To prevent a collision of notation with the chiral scalar below, write

$$V_r = (a-b)(a-c)(b-c), \quad E_B = \ell(1+3g) - g, \quad (9.4)$$

$$\Gamma_B = \ell g E_B V_r / (abc), \quad w = z + z^{-1} - 2, \quad (9.5)$$

$$C_B = \ell g \left[ \ell g t - \frac{1}{2}(g + \ell - \ell g)h \right]. \quad (9.6)$$

**Proposition 9.1** (Exact coefficients). *Writing  $\det(\xi I - J(z)) = \xi^3 - T_B \xi^2 + B_B(z)\xi - D_B(z)$ ,*

$$T_B = 3 - 6\ell - dgh, \quad (9.7)$$

$$B_B(z) = d(2 - gh) + d^2(1 - gh + g^2t) + C_B w - \frac{\Gamma_B}{2}(z - z^{-1}), \quad (9.8)$$

$$D_B(z) = (d^2 + \ell^3 w)(1 - gh + g^2t + g^3w). \quad (9.9)$$

*Proof.* Take the trace of (9.3) to obtain  $T_B$ . The middle coefficient is  $((\operatorname{tr} J)^2 - \operatorname{tr} J^2)/2$ . Collecting  $z, z^{-1}$  gives mean coefficient  $C_B$ , difference  $-\Gamma_B$ ; its value at  $z=1$  is fixed by (8.4). For the determinant, multiply  $\det(I + \ell D) = d^2 + \ell^3 w$  and  $\det H_r = 1 - gh + g^2t + g^3w$ . Similarity by  $R$  changes neither determinant. These operations derive all coefficients; Appendix D records a reproducible algebra route.  $\square$

**Theorem 9.2** (Blockwise spectral decision). *If  $\Gamma_B \sin \theta \neq 0$ , the block has non-real spectrum and the full real ring has no SPD symmetrizer. If this product vanishes, its coefficients are real and the cubic discriminant*

$$\mathfrak{D} = T_B^2 B_B^2 - 4B_B^3 - 4T_B^3 D_B - 27D_B^2 + 18T_B B_B D_B \quad (9.10)$$

*classifies simple roots: positive means three distinct real roots; negative means a real root and a conjugate pair. At zero discriminant, a Jordan test is additionally necessary.*

*Proof.*  $\operatorname{Im} B_B = -\Gamma_B \sin \theta$  whereas  $T_B, D_B$  are real. An all-real root set would have real coefficients, a contradiction. In fact a real root would obey  $(\operatorname{Im} B_B)\xi = 0$ , so if  $D_B \neq 0$  all three block roots are non-real. The conjugate block supplies their conjugates.

For a real cubic the discriminant is the product of squared root differences. It is positive for three distinct real roots; for a conjugate pair the squared difference of that pair is negative and the remaining paired product positive, giving negative sign. Repeated roots give zero. For a double root  $\rho$  and distinct  $\sigma_0$ , diagonalizability is equivalent to  $\operatorname{rank}(J - \rho I) = 1$ , or  $(J - \rho I)(J - \sigma_0 I) = 0$ . At a triple root it is equivalent to  $J$  being scalar. This follows directly from the eigenspace dimensions or minimal polynomial.  $\square$

When not all roots coincide, the repeated root at zero discriminant is

$$\rho = \frac{T_B B_B - 9D_B}{2(T_B^2 - 3B_B)}, \quad \sigma_0 = T_B - 2\rho.$$

Triple coincidence occurs when  $B_B = T_B^2/3$ ,  $D_B = T_B^3/27$ . If coefficients are non-real, repeated-root rank tests still distinguish complex Jordan defects, although SPD balance is already excluded.

The full ring admits an SPD symmetrizer exactly when every block has real semisimple spectrum. Conjugate Fourier eigenvectors then give a real eigenbasis; repeated roots in separate blocks do not create Jordan coupling.

For  $q$  at least three, an allowed phase has nonzero sine. Thus three distinct radii and

$$g\ell[\ell(1+3g) - g] \neq 0 \quad (9.11)$$

force non-real spectrum on every admitted  $N \geq 9$ , however weak the nonzero inequality. The immediate obstruction vanishes when  $g=0$ ,  $\ell = 0$ ,  $E_B = 0$ ,  $V_r = 0$ , or the particular phase is  $z = \pm 1$ . Its vanishing is not sufficient for real semisimple spectrum.

Along the weak branch,

$$\Gamma_B = \ell g E_B \left( \frac{\varepsilon}{2\varepsilon k + 3g} \right)^3 V(\kappa) \eta^3 + O(\eta^4).$$

At a simple nonzero equal-background block root  $\rho$ , differentiating the characteristic equation gives

$$\text{Im } \delta\rho = \frac{\Gamma_B \sin \theta \rho}{\prod_{\sigma' \neq \rho} (\rho - \sigma')} + O(\eta^4).$$

Repeated or zero baseline roots require the exact polynomial tests instead. Complex roots can remain strictly inside the unit disk. They are oscillatory linear-response modes, with no physical wave interpretation. *Checkpoint authority: GR2 §5, L21–L24.*

## 10 Six-ring reality, defect and complex unfolding

For  $N=6$  only  $z = 1, -1$  occur;  $N=12$  adds  $z = i, -i$ . The first block is the triad. At  $z=-1$  the polynomial is real with  $w=-4$ ; at  $z = \pm i$ , its imaginary middle coefficient is  $\mp \Gamma_B$ . Thus the same native background can have a real-diagonalizable six-ring response and a non-real twelve-ring response.

At equal radii the antiperiodic block has double root  $\rho_1 = (1 - \ell)(1 - g)$  and simple root  $\rho_4 = (1 - 4\ell)(1 - 4g)$ , with gap  $3E_\pi$ ,  $E_\pi = g + \ell - 5g\ell$ . For  $v_i = u_i/\sqrt{k}$ , its discriminant is

$$\mathfrak{D}_\pi = 6g^2(1 - \ell)^2 \sum_i v_i^2 (3E_\pi)^4 \eta^2 + O(\eta^3). \quad (10.1)$$

Indeed the first derivative is  $(I + \ell D_\pi)g([D_\pi, \text{diag } v] + 3 \text{diag } v)$ . Its projection onto the double  $-1$  eigenspace of  $D_\pi$  is  $3g(1 - \ell)\Pi_\pi \text{diag}(v)\Pi_\pi$ , symmetric with the squared gap shown in (10.1). Multiply by the fourth power of separation from the third root. At a fixed stable baseline with  $g(1 - \ell)E_\pi \neq 0$ , weak nontrivial inequality therefore gives real simple antiperiodic roots and an SPD full-ring metric, despite the diagonal obstruction.

The neighbourhood is not uniform near exceptional baselines. At  $E_\pi = 0$ ,  $\ell = g/(5g - 1)$ , the first variation  $K_\pi$  has trace and determinant zero and

$$\text{tr } K_\pi^2 = \frac{3g^2(4g - 1)^2}{(5g - 1)^2} \sum_i v_i^2.$$

Unless this coefficient vanishes, its three first-order roots are distinct and real. At  $\ell = 1$ , the antiperiodic synchronizer has rank one. The exact response can instead be defective.

**Theorem 10.1** (Exact native nilpotent response and unfolding). *Take*

$$r = (1 - \tau, 1, 1 + \tau), \quad g = (1 - \tau^2)/4, \quad \ell = 1, \quad \varepsilon = 1/20, \quad k_i = r_i^2 - \frac{g(\Delta_{3r})_i}{\varepsilon r_i}, \quad (10.2)$$

with  $0 < |\tau|$  sufficiently small. Then

$$J(-1) = \frac{\tau^2}{4} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 + \tau & 2 & 1 - \tau \end{pmatrix}. \quad (10.3)$$

It has rank one, square zero, and Jordan sizes  $2 + 1$  at eigenvalue zero. Holding  $r, g, k_i, \varepsilon$  fixed and varying only  $\ell = 1 + \delta$ , its discriminant satisfies

$$\mathfrak{D}_\pi = \frac{27}{2} \tau^{12} \delta^3 + O(\delta^4). \quad (10.4)$$

For each fixed nonzero small tau, sufficiently small negative delta gives a complex pair, positive delta gives three distinct real roots, and zero delta gives the defect.

*Proof.* Substitute (10.2) in (9.3); direct multiplication gives (10.3). Its nonzero row and column pair to zero, proving rank one and square zero. A three-dimensional nonzero square-zero rank-one operator has one size-two block and one size-one block. The coefficients under the stated variation are exactly

$$T_B = -3\delta/2, \quad (10.5)$$

$$B_B = -\frac{3\delta}{16} [8\tau^4 + \delta(10\tau^4 + \tau^2 - 3)], \quad (10.6)$$

$$D_B = \frac{\delta^2 \tau^4 (4\delta + 3)(9 - \tau^2)}{16}. \quad (10.7)$$

Substituting into (9.10) leaves the cubic coefficient  $27\tau^{12}/2 > 0$ , proving the sign conclusions and simplicity away from zero for small delta.  $\square$

This is a native fixed point by construction, not an arbitrary matrix. The coefficients are positive for small tau, with mean  $k = 1 - 28\tau^2/3$ . For each fixture,  $\eta = \tau$  and  $\kappa_i = (k_i - k)/\tau$  put it in the admitted form. Varying tau in this auxiliary family does *not* keep kappa fixed. At the rational fixture  $\tau = 10^{-3}$ , exact contraction-box inequalities connect the point to the unique weak branch for its fixed direction (Appendix E).

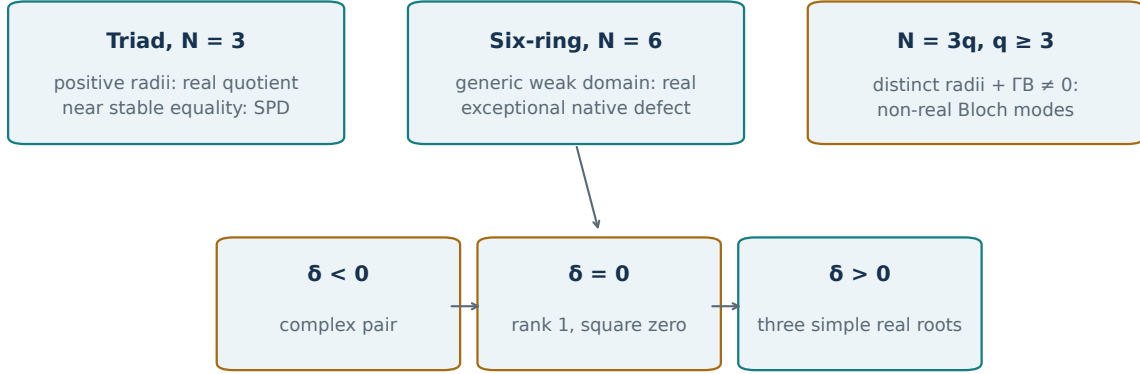
The equal-reference noncommon phase multipliers at  $\ell = 1$  are  $0, -1/2 - 3\tau^2/2, -3\tau^2$ ; its radial multipliers are strictly inside the unit disk for small tau. At the unequal point the triad transverse roots are  $-\frac{1}{2}(1 \pm \tau\sqrt{3 + 6\tau^2})$ , and the antiperiodic eigenvalues vanish. Moreover  $\|J(-1)\|_\infty = \tau^2$ . Thus the defect and its sufficiently small unfolding can remain normally stable. Stability, reality and diagonalizability are separate questions. *Checkpoint authority: GR2 §6, L25–L27.*

## 11 Equivariant spatial observables

The shell selects a polar vertical  $e_z$ , outward normals  $n_i$ , tangents  $t_i = e_z \times n_i$  and centre  $o$ . For the triad write

$$\Omega = q + ip, \quad e = (1, 1, 1)^T, \quad q = ae + u, \quad p = be + v, \quad u, v \perp e, \\ N_f = [n_A \ n_B \ n_C], \quad T_f = [t_A \ t_B \ t_C].$$

## Finite-ring spectral alternatives



Derivation: GR2 §§3-6. Bottom row: fixed nonzero small  $\tau$ , sufficiently small  $\delta$ ; stability can persist.

Figure 3: Finite spectral hierarchy, with hypotheses stated on each branch. The six-ring three-way alternative uses (10.2) and variation of the existing synchronizer parameter only. The larger-ring statement requires (9.11) and distinct radii. All are finite linear-response results. Source: GR2 §§3–6.

Exact coordinates appear in Appendix A. The frame matrices have Gram matrix  $(3/2)\Pi$ ,  $\Pi = I - ee^T/3$ .  $N_f$  is not ring size.

Let P cycle A,B,C and  $Q_v$  swap A,C. Polar covariance of the adopted decoder gives the actual state actions

$$C_3 : (q, p) \mapsto (Pq, Pp), \quad \sigma_v : (q, p) \mapsto (-Q_v q, Q_v p), \quad \sigma_h : (q, p) \mapsto (q, -p). \quad (11.1)$$

They are not ordinary permutation of both channel triples. Polar free vectors transform by G, axial vectors by  $\det(G)G$ , and points affinely about o. This is the kinematic representation at every state. Dynamical covariance carries the coefficient triple k with the face permutation; at fixed unequal k only its stabilizer remains. Mirrors involving common phase require the regular pre-synchronization domain for nonzero lambda. The census does not assume a global dynamical symmetry across the zero stratum.

**Theorem 11.1** (Complete ambient census). *For real polynomial state maps equivariant under full  $D_{3h}$ , the linear polar basis is  $T_f u, be_z$ , and the homogeneous quadratic polar basis is*

$$aN_f u, \quad N_f(u \odot u), \quad bN_f v, \quad N_f(v \odot v), \quad \Gamma_{ch} e_z, \quad \Gamma_{ch} = e^T(u \times v). \quad (11.2)$$

*The axial linear basis is  $T_f v, ae_z$ , and the quadratic axial basis is  $aN_f v, bN_f u, N_f(u \odot v)$ . No nonzero constant ambient equivariant exists.*

*Proof.* The state representation on a,u,b,v decomposes as  $A'_2 \oplus E' \oplus A''_2 \oplus E''$ . The polar target is  $E' \oplus A''_2$ , the axial target  $E'' \oplus A'_2$ . Direct expansion of q-q, p-p and q-p products gives

$$\text{Sym}^2 \mathcal{X}_{\mathbb{R}} = 4A'_1 \oplus 4E' \oplus 2A''_1 \oplus A''_2 \oplus 3E''. \quad (11.3)$$

Each unmixed sector gives two invariant scalars and two transverse copies; the mixed sector gives two mirror-odd scalars, the oriented scalar  $\Gamma_{ch}$ , and three transverse copies. Counting matching irreducibles gives dimensions 2,5 and 2,3. The listed maps obey the generator laws. Independent means and distinct q-q, p-p and q-p monomials prove independence, hence completeness. Neither target contains the trivial representation, so constant maps vanish.  $\square$

An additional demand of common-phase invariance still leaves the three-dimensional quadratic polar space

$$\text{span}\{N_f(au + bv), N_f(u \odot u + v \odot v), \Gamma_{\text{ch}}e_z\}. \quad (11.4)$$

The axial invariant space is  $\text{span}\{W\}$ ,  $W = T_f(q \times p)$ , and no invariant linear map survives. Apply the common-phase generator  $\sum_i(-p_i\partial_{q_i} + q_i\partial_{p_i})$  to the complete bases: its kernel is exactly the displayed dot-product and oriented-area combinations. This is an optional kinematic condition, not a global phase quotient of  $F$  at zero.

### 11.1 Chiral projections and native observable changes

$$C_{\text{ch}} = q \times p = e \times (av - bu) + u \times v, \quad W = \sqrt{3}(bN_fu - aN_fv), \quad \Gamma_{\text{ch}} = e^T C_{\text{ch}}. \quad (11.5)$$

Raw  $C$  transforms as  $P, Q_v, -I$ , neither ambient polar nor axial law. Its transverse part gives horizontal axial  $W$  and its common part gives vertical polar  $\Gamma_{\text{ch}}e_z$ . Each is unique up to scale within its corresponding antisymmetric sector, not among all observables. Together

$$C_{\text{ch}} = \frac{2}{3}T_f^T W + \frac{1}{3}\Gamma_{\text{ch}}e.$$

For pure transverse independent  $u, v$ ,  $W$  vanishes but  $\Gamma$  does not. Axial parity alone need not mean chirality:  $ae_z$  is nonzero on common purely real states.

Equal channels kill every quadratic spatial map. Unequal amplitudes at common phase can yield intensity/coherence vectors while  $C_{\text{ch}} = 0$ . At  $\Omega = r_0e + iv$ ,  $v \perp e$ , the first-order axial readout is  $-\sqrt{3}r_0N_fv$ . For  $u = (1, -1, 0), v = (1, 1, -2), a = b = 0$ , one has  $C_{\text{ch}} = 2e, \Gamma_{\text{ch}} = 6, W = 0$ .

An observable  $V$  receives no additional evolution:

$$\Delta V = V(F(\Omega)) - V(\Omega).$$

For instance  $\Delta(T_fu) = T_f\Delta u$  and

$$\Delta\Gamma_{\text{ch}} = e^T(u \times \Delta v + \Delta u \times v + \Delta u \times \Delta v).$$

All quadratic increments follow substitution; local response is  $DV(\Omega_*)DF(\Omega_*)$ . At a positive equal triad, radial mean, radial transverse, imaginary mean and imaginary transverse factors are  $1 - 2\epsilon k$ ,  $1 - 2\epsilon k - 3g$ ,  $1$ ,  $(1 - 3g)(1 - 9\lambda)$ . On unequal branches use (5.4). Existing one-step examples generate basis observables from zero (Appendix H); nonzero readouts at fixed states further show why a vector value is not displacement per step.

Symmetry therefore does not select a unique state-dependent core position or motion observable. A physical position law remains an additional ingredient. *Checkpoint authority: CM0 §§2–6, L28–L33.*

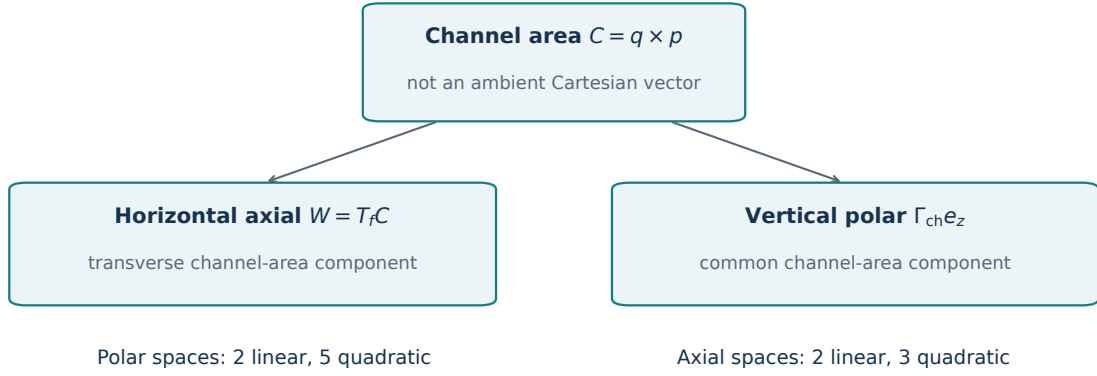
## 12 State-shell attachment and finite-carrier multiplicity

The accepted fixed mesh has three octagonal faces, three vertical seams and two nine-edge rim cycles. A face lies in the affine plane  $c_i + W_i$ ,  $W_i = n_i^\perp$ ;  $W_i$  is unbounded.

**Definition 12.1** (Adopted decoder).

$$D_i(q_i + ip_i) = q_it_i + p_ie_z, \quad E_i(v_i) = t_i \cdot v_i + ie_z \cdot v_i, \quad D : \mathbb{C}^3 \rightarrow \bigoplus_i W_i. \quad (12.1)$$

## Spatial covariance and independent chiral projections



Derivation: CM0 §§3–4. Dimensions count maps; parity does not supply a physical position or axis.

Figure 4: Independent chiral projections and the larger equivariant spaces. Axial denotes reflection parity, not vertical direction. Multiplicities count polynomial-map spaces, not physical modes. Source: CM0 §§3–4 and (11.2)–(11.5).

Orthonormality proves  $ED = I$ ,  $DE = \text{diag}(I - n_i n_i^T)$  on ambient arrays, and  $\sum \|D_i \Omega_i\|^2 = \sum |\Omega_i|^2$ . Basing at the adopted centre gives  $(c_i, v_i)$ ; the endpoint  $c_i + v_i$  need not lie in the finite face and is not a particle-position law.

**Proposition 12.2** (Discrete connection and conjugacy). *The maps  $T_{ij} = t_i t_j^T + e_z e_z^T : W_j \rightarrow W_i$  are orthogonal isomorphisms satisfying  $T_{ij} T_{jk} = T_{ik}$ . They define a flat discrete rank-two connection on the face-label graph. The exact conjugate update is*

$$\tilde{v}_i = v_i + \varepsilon(k_i - \|v_i\|^2)v_i + g \sum_{j \neq i} (T_{ij} v_j - v_i), \quad (12.2)$$

$$v_i^+ = \cos \delta_i \tilde{v}_i + \sin \delta_i n_i \times \tilde{v}_i, \quad F_{\text{face}} = DFE. \quad (12.3)$$

*Proof.* T carries the two frame coefficients unchanged, proving composition and isometry. Closed transport is identity on its tangent domain, a projector ambiently. Encoding the first line recovers the native pre-stage. The identity  $E_i(n_i \times v) = iE_i(v)$  turns the second into phase multiplication, including the zero value. D/E invert on their domain.  $\square$

The implementation steps stored canonical Omega and then decodes; it does not repeatedly encode rounded displays. No second recurrence or carrier motion is added.

A section on three labelled fibres differs from a field on each continuous polygon, seam/rim traces, shell deformation and physical position. Only the first is natively defined. Conditional constant face extensions illustrate the distinction: ambient equality at every seam forces  $q = 0, p = be$ , because distinct tangent planes intersect only in  $\mathbb{R}e_z$ . Matched parallelism instead forces  $q = ae, p = be$ . Neither criterion is imposed on arbitrary native states.

## 12.1 Multiplicity on finite carriers

| Carrier                                    | degree 0 | degree 1 | degree 2 |
|--------------------------------------------|----------|----------|----------|
| Ordinary face scalar or normal coefficient | 1        | 1        | 8        |
| Polar tangent vector per face              | 0        | 4        | 8        |
| Polar ambient vector per face              | 1        | 5        | 16       |
| Polar centre vector                        | 0        | 2        | 5        |
| Axial centre vector                        | 0        | 2        | 3        |
| Ordinary scalar per unordered seam         | 1        | 1        | 8        |
| Polar vertical tangent vector per seam     | 0        | 2        | 4        |
| Polar ambient vector per unordered seam    | 1        | 5        | 16       |
| Scalar per whole rim                       | 1        | 1        | 5        |
| Polar ambient vector per whole rim         | 1        | 3        | 12       |

The proof uses (11.3) and the target representations; complete independent bases appear in Appendix G. Each seam is indexed equivariantly by its opposite face. Horizontal reflection exchanges the two rims. No individual seam or boundary point is selected.

Already linear tangent assignments have four independent constants:

$$Q = \alpha ae + \beta u, \quad P = \gamma be + \delta v, \quad v_i^{\text{cand}} = Q_i t_i + P_i e_z.$$

D adopts all four as one. A normal coefficient  $f_i n_i$  does not specify compatible vertex displacement; a whole-loop vector need not be tangent to every segment. Pointwise fields need a spatial function space beyond this census.

## 12.2 Preparation and boundary types

The exact type chain is

$$(R, d; \xi \in \mathbb{C}^2) \mapsto \theta, \psi_0 \in \mathbb{C}^2 \otimes \mathbb{C}^3 \xrightarrow{U^n} \psi_n \xrightarrow{\chi^\dagger \otimes I} \Omega_0 \xrightarrow{D} \bigoplus_i W_i. \quad (12.4)$$

The named toy response uses

$$\theta = \arccos(d/2R), \quad G(\theta) = \sqrt{(2\theta - \sin 2\theta)/\pi}, \quad \psi_0 = G(\theta)(\xi \otimes f_0), \quad f_0 = e/\sqrt{3},$$

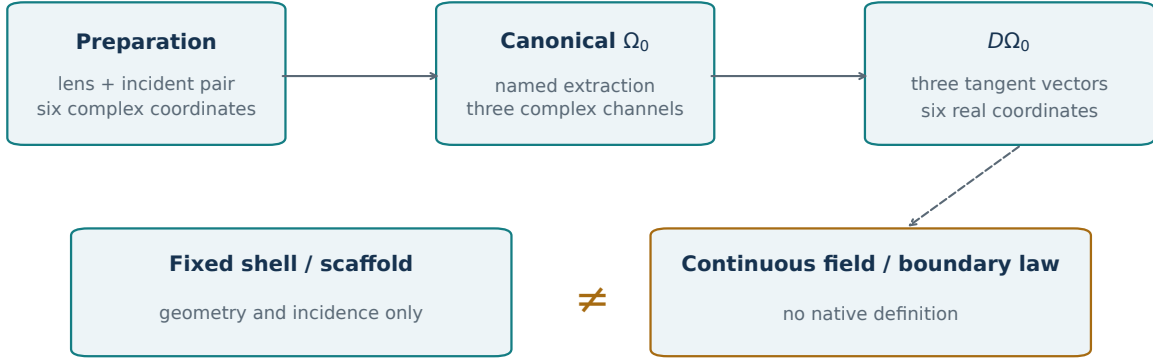
with  $R > 0, 0 \leq d \leq 2R$ . The fixed transfer is  $U = B_s \otimes A_s$ , with named normalized eigenbra  $\chi^\dagger B_s = b_\chi \chi^\dagger$ . Its established reduction is

$$\Omega_0 = G(\theta)(\chi^\dagger \xi) b_\chi^n h_n f_j, \quad h_n = (r_s q_s^2 c_s^2)^m (1, q_s, q_s^2 c_s)_j, \quad n = 3m + j,$$

where  $q_s = e^{-.423}, r_s = e^{.577}, c_s = 1 - .618$  are fixed initialization literals;  $f_j$  are the ordered Fourier vectors.

The final Omega can use D. No map attaches preparation internals to material rims, seams, faces or scaffold cells. The complex six-state space has twelve real dimensions; the incident pair is not two rim traces; the lens chart is not computed from shell geometry. Source signatures and imports use them for initialization only; F receives Omega and its native parameters. The reference scaffold is a separate fixed family of closed sets with a special explicit Paper-C comparison, not a state attachment to gaps. There is no native boundary/seam feedback. *Checkpoint authority: SA0 §§2–7; prior Paper B/Atlas 04 and Atlas 08, L34–L37.*

## Typed state-to-shell attachment



Derivation: SA0 §§2-3,7. Dashed arrow is absent, not a proposed model extension.

Figure 5: Typed attachment. D is the existing invertible representation; preparation reaches it through final Omega. Dashed absent arrows mark missing definitions, not proposed interactions. Source: SA0 §§2-3,7 and (12.4).

## 13 Closure hierarchy and coherence quotient

**Theorem 13.1** (Fibre closure criterion). *For maps  $\mathcal{A} : X \rightarrow Y$ ,  $F : X \rightarrow X$ , an induced map satisfying  $\mathcal{A}F = F_{\mathcal{A}}\mathcal{A}$  on  $\mathcal{A}(X)$  exists iff*

$$\mathcal{A}(x) = \mathcal{A}(x') \implies \mathcal{A}(Fx) = \mathcal{A}(Fx'). \quad (13.1)$$

*Proof.* Necessity is composition with  $F_{\mathcal{A}}$ . For sufficiency define  $F_{\mathcal{A}}(y) = \mathcal{A}(Fx)$  using any representative; the implication makes it well defined.  $\square$

This asserts no locality or smoothness. A restricted-domain claim requires the same implication on every fibre in that domain.

The complete decoded/based tangent triple closes globally by bijectivity of D. One face generally does not: with  $\varepsilon = 1/20, g = 1/5, k = e$ , real inputs  $(1, 1, 1)$  and  $(1, 2, 1)$  have equal A coefficient but next coefficients 1 and  $6/5$ . Pre-states are positive, so this holds for all lambda.

### 13.1 Lower-readout obstructions

At the same eps,g,k,  $(1, 1, 1)$  and  $(1, -1, 1)$  have identical intensities but next intensities

$$(1, 1, 1) \quad \text{and} \quad (9/25, 1/25, 9/25).$$

Synchronization preserves moduli; total intensity also differs. At  $g=0$  the explicit exception is  $I_i^+ = I_i[1 + \varepsilon(k_i - I_i)]^2$ .

For chirality, set lambda=0 and compare

$$(q, p) = (e, (1, -1, 0)), \quad (q', p') = (2e, (1, -1, 0)/2).$$

Both give initial  $C_{\text{ch}} = (1, 1, -2)$ , but next values

$$(7/20, 7/20, -133/200), \quad (323/1600, 323/1600, -1273/3200).$$

Their common and transverse components differ.  $C$ ,  $W$ ,  $\Gamma_{\text{ch}}$  and their joint chiral representation therefore fail general closure. These exact fibre counterexamples do not exclude special parameter or invariant-subset exceptions.

### 13.2 Complex coherence

Distinguish the response matrix  $H_r$  from the coherence

$$\mathcal{H} = \Omega\Omega^\dagger = S_G - iA_G, \quad S_G = qq^T + pp^T, \quad A_G = qp^T - pq^T.$$

For rank-at-most-one positive semidefinite Hermitian  $\mathcal{H}$ , put

$$M(\mathcal{H}) = I + \varepsilon \text{diag}(k - \text{diag } \mathcal{H}) + g\Delta_3, \quad \tilde{\mathcal{H}} = M\mathcal{H}M^T. \quad (13.2)$$

**Theorem 13.2** (Regular complex-coherence closure). *Where all  $\tilde{\mathcal{H}}_{ii} > 0$ , set*

$$\rho_i = \sqrt{\tilde{\mathcal{H}}_{ii}}, \quad \delta_i = \lambda \sum_{j \neq i} \text{Im} \left( \frac{\tilde{\mathcal{H}}_{ji}}{\rho_j \rho_i} \right)^3, \quad U_\delta = \text{diag}(e^{i\delta_i}).$$

The exact induced update is

$$\boxed{\mathcal{H}^+ = U_\delta \tilde{\mathcal{H}} U_\delta^\dagger}. \quad (13.3)$$

*Proof.*  $M\Omega$  is the native pre-state. Its normalized ji coherence entry is  $e^{i(\tilde{\phi}_j - \tilde{\phi}_i)}$ . Cubing supplies the native sine increment. The final phase multiplication gives (13.3); everything depends only on  $\mathcal{H}$ . Equivalently its fibres are common-phase orbits, and  $F$  is common-phase equivariant on this domain.  $\square$

Iteration is justified while the regularity condition persists; the domain is not claimed invariant.

**Theorem 13.3** (Regular real-Gram closure). *The real Gram matrix  $S_G \succeq 0$ , of rank at most two, also closes where  $(MS_G M^T)_{ii} > 0$ .*

Here  $M = I + \varepsilon \text{diag}(k - \text{diag } S_G) + g\Delta_3$ ; its dependence is only on the retained diagonal.

*Proof.* Equal real Gram factors  $QQ^T = Q'Q'^T$  have identical pairwise row inner products. The row-span identification is an isometry and extends to an  $O(2)$  transformation, giving  $Q' = QO$ . Such changes are common phase and/or conjugation of  $\Omega$ . The real-parameter  $F$  is conjugation equivariant, and on this domain common-phase equivariant. Output real Grams therefore agree, proving the fibre criterion, including rank-deficient factors.  $\square$

$S$  loses signed orientation; closure survives because the discarded distinction is a symmetry. This is a nonlocal channel quotient, not one scalar per face.

### 13.3 The essential zero-stratum exception

At  $\varepsilon = g = 0, \lambda = 1/10$ , let  $z = (1, -1, 0)$ ,  $z' = e^{i\pi/6}z$ . They have identical complex and real Grams. For  $z$  all sine increments vanish; for  $z'$  the two nonzero increments are  $-\lambda, +\lambda$ , since the zero phase remains zero. Thus

$$\mathcal{H}_{12}^+ = -1, \quad (\mathcal{H}')_{12}^+ = -e^{-i/5}, \quad S_{G,12}^+ = -1, \quad (S'_G)_{12}^+ = -\cos(1/5). \quad (13.4)$$

They differ since  $0 < 1/5 < \pi$ . Neither quotient closes globally for the native nonzero-lambda convention. Both congruence updates are global when lambda=0. No repaired convention is proposed.

The hierarchy is exact: D retains full state; lower face/intensity/area readouts generally discard necessary information; coherence admits a regular symmetry quotient; the native zero convention prevents unrestricted extension. Fixed geometry is trivially constant; preparation receipts, clocks and observer memory are other data types, not autonomous Omega-only shell variables. *Checkpoint authority: SA0 §8, L38–L42.*

## 14 Interpretation boundaries

The following distinctions are part of the mathematical result, not qualifications to be removed by a more suggestive picture.

| Established mathematical object |        | Unsupported identification   |
|---------------------------------|--------|------------------------------|
| Response inner product          | $\neq$ | Spacetime metric             |
| Screened graph response         | $\neq$ | Newtonian gravity            |
| Complex response eigenmode      | $\neq$ | Gravitational wave           |
| Non-reversible cycle imbalance  | $\neq$ | Physical current             |
| SPD symmetrizer                 | $\neq$ | Privileged physical geometry |
| Tangent vector                  | $\neq$ | Physical position            |
| Spatial observable              | $\neq$ | Core trajectory              |
| Axial parity                    | $\neq$ | Physical rotation axis       |
| State-to-face representation    | $\neq$ | Material deformation         |
| Graph coupling                  | $\neq$ | Seam flux                    |
| Lens/SRG preparation            | $\neq$ | Boundary dynamics            |

The inequality signs denote unestablished identifications, not an assertion that no future model could relate the objects. The response inner product depends on a state background; the panel metric and matched connection are fixed. Their different domains prevent an automatic identification. Likewise, the non-real eigenvalues are statements about powers of a finite Jacobian, without an assigned physical clock or wavelength.

For nonnegative response entries, a row-sum-one matrix has a mathematical finite-state Markov interpretation, and the cycle obstruction tests reversibility. Without entrywise nonnegativity the algebraic balance question still makes sense, but probability currents should not be asserted. Even a valid Markov interpretation would not itself establish a physical current.

The names of the GR checkpoints preserve the original investigation question. Their conclusion is response geometry and finite spectral structure, not a physical gravity theory. No new physical model, kernel integration, historical-motion import or parked-lane derivation is part of this paper.

## 15 Relation to established mathematics

Equation (5.3) belongs to the weighted graph Dirichlet-form setting described by Chung [15, Chapter 1]. The reversible/non-reversible distinction is an edge-balance problem; in a positive-entry stochastic domain, the triangle proof is the finite Kolmogorov cycle condition. Detailed balance and reversibility are standard Markov-chain concepts [14, §1.6]; the matrix cycle condition is also treated in [19, §4]. The native cycle factor and its exceptional parameter loci are derived here from the checkpoint equations, not imported from these references.

The terminology in Section 7 separates this node-local condition from arbitrary SPD self-adjointness. McKee and Smyth [19, Definition 4.1] use “symmetrizable” for similarity to a symmetric matrix by a positive diagonal change of basis. Theorem 7.1, proved here, permits a general SPD inner product and is correspondingly weaker than diagonal balance.

The exact translation-by-three decomposition is finite Fourier or Bloch analysis. Kuchment’s overview [16, §4] describes Floquet decomposition in the broader theory of periodic operators, including operators on graphs. Our finite blocks are derived directly; that context supplies neither a continuum approximation nor a physical wavelength for the native response.

The matched maps identify specified vector fibres compatibly. Graph connection and vector-fibre terminology has an established setting in Singer and Wu [17]. This comparison does not identify the native transport with their data-derived connection Laplacian or infer an ambient gravitational connection. Likewise, the observable census uses finite-dimensional equivariant polynomial representation theory; Golubitsky, Stewart and Schaeffer [18] provide general group-theoretic and equivariant-map context. The native representation, multiplicities and observable formulas remain derived in this paper and its cited checkpoints.

The screened response is a finite modified-Helmholtz inverse. The closure theorem is the descent-to-a-quotient criterion, proved here for the native recurrence with its zero convention. These comparisons assign mathematical types; they assert neither novelty nor a physical dictionary. The internal accepted bibliography supports the native definitions and prior ownership [6, 7, 8, 9, 10, 11, 12, 13]. The six external references supply targeted surrounding context only. All model-specific claims and the linear-algebra, Fourier, cycle and quotient arguments needed here remain self-contained; no priority claim is made.

## 16 Limitations, open questions and conclusion

Every nonlinear branch existence and attraction statement here is local for a chosen finite ring unless explicitly stated otherwise. The positive-branch algebra can have a broader domain than its local stability consequences. The generic six-ring neighbourhood is not uniform near the nilpotent fixture. The cubic imaginary-root expansion requires a simple nonzero baseline root; exact polynomial and rank tests cover the exceptional cases.

The finite-carrier census fixes the full  $D_{3h}$  representation and polynomial degree. It does not classify arbitrary nonpolynomial maps, all continuous shell fields or observations depending jointly on evolving additional geometry. The coherence quotient is regular-domain mathematics, with the native zero-stratum counterexample retained. No proof of a complete nonlinear continuum limit, global positive-branch continuation, canonical dense response metric or physical position selection is supplied.

One bounded open problem already identified by SA0 is the minimal subset of existing pairwise coherences that, together with face intensities, supports autonomous regular-domain closure. Intensity is insufficient; full real Gram is sufficient. This synthesis does not begin that calculation.

The established chain is nevertheless substantial. Background inequality selects exact pre-stage weights and conductances; synchronization generically removes diagonal reversibility. The spectral consequence depends on the finite sector: local positive triads permit a non-diagonal SPD metric; for  $N \geq 9$ , three-distinct repeated backgrounds satisfying the stated nondegeneracy condition force non-real modes; and a six-ring admits a native square-zero defect with a sign-resolved unfolding. Independently, the spatial representation has multiple equivariants, while its adopted complete face view is lossless. The coherence quotient explains a specific level at which passive information can evolve autonomously and identifies exactly why the native zero

convention blocks the unrestricted statement. These results describe the existing mathematical map without a new physical premise.

## 17 Reproducibility, evidence and availability

The source checkpoint is

`da461294d724997bd7033c7cea9cc9a0b5f5031d.`

The scientific source checkout, existing papers, Atlas and all fifteen GR0/GR1/GR2/CM0/SA0 artifacts are preserved. Only this new publication lane is authored. Source hashes, the initial dirty status, final preservation comparison, theorem ledger and paper-source crosswalk are included in the package. Private machine locators occur only in provenance records, not in scientific statements.

Retained counts are

| Checkpoint          | GR0 | GR1 | GR2 | CM0 | SA0 | total |
|---------------------|-----|-----|-----|-----|-----|-------|
| Exact/source checks | 45  | 38  | 42  | 78  | 226 | 429   |

These heterogeneous predicates include symbolic identities, matrix-entry groups, source-dependency checks and bounded structural witnesses. They are not 429 theorems. Written arguments are the proof authority. Predecessor scripts are referenced, not rerun into their result files; their verification records are retained with unchanged source hashes.

Numerical evidence is kept by method and quantity. Native complex128 finite-difference errors are not combined with 100-digit transcription errors, branch residuals, characteristic residuals, metric balance residuals or spectral imaginary parts. In particular, a native Jacobian residual does not resolve imaginary roots smaller than that residual. Appendix I summarizes these separate measurements; the complete unaggregated records are in the companion retained-evidence JSON.

Paper-local verification independently checks displayed factorization, polynomial, finite-block, nilpotent and quotient identities and validates manuscript/source references. Its count and results are separate from 429. Figure sources generate five vector schematics with explicit derivation notes. Build commands, a proposed file allowlist, publication text and three optional public-post formulations are supplied; none is published automatically. Research publication v0.1 carries no claim of journal submission, external peer review, a repository release or DOI registration.

**Preparation disclosure.** The manuscript was assembled with AI-assisted mathematical writing, source reconciliation, symbolic verification and typesetting. Scientific review, author approval and any publication decision remain with the project owner. No claim of independent peer review is made.

## A Notation and conventions

| Symbol                                        | Meaning                                                                                                                      |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| $N = 3q, \Delta_N$                            | Admitted ring length and native negative cycle Laplacian                                                                     |
| $\ell = 3\lambda$                             | Abbreviation for existing synchronizer strength                                                                              |
| $r, R, H_r, P, S_\phi, J_\phi$                | Positive radii, diagonal radii, symmetric pre-response, phase pre-response, synchronizer derivative, complete phase response |
| $h, t, d$                                     | Triad scale-invariant algebraic quantities and $d = 1 - 3\ell$ ; $t$ is not time                                             |
| $\Gamma_B$                                    | Bloch imaginary-coefficient obstruction                                                                                      |
| $q, p = \text{Re } \Omega, \text{Im } \Omega$ | Real channel vectors; not ambient components                                                                                 |
| $a, b, u, v$                                  | Their means and transverse components in the spatial-observable sections                                                     |
| $N_f, T_f$                                    | Fixed normal/tangent frame matrices; not graph size or physical time                                                         |
| $C_{\text{ch}}, W, \Gamma_{\text{ch}}$        | Raw channel area and its horizontal-axial/vertical-polar projections                                                         |
| $\mathcal{H}, S_G, A_G$                       | Complex coherence, real Gram and antisymmetric pair-area matrix                                                              |

In the spectral sections  $a, b, c$  denote positive radii; in the observable sections  $a, b$  are channel means. The contexts and subscripts prevent an identification. The full mirror action is stated in (11.1); the degree counts are not merely  $C_3$  counts.

The exact shell frames, with  $o = (0, \sqrt{3}/6, 0)$ , are

| $i$ | $c_i$                   | $n_i$                   | $t_i$                    |
|-----|-------------------------|-------------------------|--------------------------|
| $A$ | $(-1/4, \sqrt{3}/4, 0)$ | $(-\sqrt{3}/2, 1/2, 0)$ | $(-1/2, -\sqrt{3}/2, 0)$ |
| $B$ | $(0, 0, 0)$             | $(0, -1, 0)$            | $(1, 0, 0)$              |
| $C$ | $(1/4, \sqrt{3}/4, 0)$  | $(\sqrt{3}/2, 1/2, 0)$  | $(-1/2, \sqrt{3}/2, 0)$  |

Direct dot products prove  $N_f^T N_f = T_f^T T_f = (3/2)\text{II}$ . At a general real fold angle  $\beta$ , the accepted panel maps have material derivatives

$$\partial_u X_1 = (-\cos \beta, -\sin \beta, 0), \quad \partial_u X_2 = (1, 0, 0), \quad \partial_u X_3 = (-\cos \beta, \sin \beta, 0), \quad \partial_z X_i = e_z.$$

Each horizontal derivative has unit norm and is orthogonal to  $e_z$ ; their pullback metrics are therefore  $I_2$ . This local identity does not assert an embedded closed assembly at arbitrary fold angle. Embedding folds do not imply a variable intrinsic panel metric.

For completeness, the preparation Fourier vectors are  $f_0 = e/\sqrt{3}$ ,  $f_1 = (1, \bar{\omega}, \omega)^T/\sqrt{3}$ ,  $f_2 = \bar{f}_1$ ,  $\omega = e^{2\pi i/3}$ . With  $p_0 = f_0 f_0^\dagger$ ,

$$A_s = [q_s I + (r_s - q_s) p_0] \text{diag}(1, \bar{\omega}, \omega) [c_s I + (1 - c_s) p_0].$$

The fixed two-factor matrix is

$$B_s = \text{diag}(e^{-i\alpha}, e^{i\alpha}) \begin{pmatrix} \cos \beta & -i \sin \beta \\ -i \sin \beta & \cos \beta \end{pmatrix}, \quad \alpha = 2\pi(.244), \quad \beta = \pi(.244).$$

This fully specifies the algebra used in the type chain, without assigning material boundary meaning.

## B Complete theorem and claim ledger

This is the anti-overstatement ledger. Predicate ordinals resolve to the named lists in `verification/retained_predicates.json`; checks do not replace proof locations.

## L01 Native map and regular chart

**DEFINITION / ADOPTION.** The native map is amplitude/coupling followed by harmonic-three phase synchronization;  $\text{Arg0}(0)=0$ .  $N=3q$ , coefficients repeated every three nodes.

**Domain:** Real parameters; finite states; differential claims only on nonzero pre-sync charts.

**Authority/proof:** GR0 §5.1–5.2; native inventory §3; main text §2.

**Prior ownership:** Yes: Paper A §6.1

**Qualification:** No global differentiability at zeros for nonzero lambda.

**Interpretation limit:** No physical time or spatial mesh is introduced.

**Supporting predicates:** GR0 ordinals 33–35

## L02 Panel and state isometries

**DERIVED IDENTITY.** The panel pullback metric is  $I_2$ . The decoder  $D_i(q + ip) = qt_i + pe_z$  is isometric;  $ED = I$ , while  $DE$  is the ambient tangent projector.

**Domain:** Canonical frames; arbitrary real fold for individual panel isometry; D/E restricted to tangent domain.

**Authority/proof:** GR0 §4(a); main text §12.

**Prior ownership:** Yes: Papers B/C; Atlas 03/04

**Qualification:** DE is not the ambient nine-dimensional identity.

**Interpretation limit:** Fixed panel geometry and state geometry remain distinct.

**Supporting predicates:** GR0 ordinals 1, 3, 5, 7–8, 12–13, 17–18

## L03 Matched transport

**EXACT THEOREM.**  $T_{ij} = t_i t_j^T + e_z e_z^T$  satisfies  $T_{ij} T_{jk} = T_{ik}$ . Closed products restrict to identity on  $W_i$ .

**Domain:** Labelled tangent fibres; ambient maps have rank two.

**Authority/proof:** GR0 §4(b)–(c); SA0 §3.2; main text §12.

**Prior ownership:** Yes: Paper B Proposition 2

**Qualification:** Ambient closed product is a projector.

**Interpretation limit:** No arbitrary surface-path or physical connection is derived.

**Supporting predicates:** GR0 ordinals 9–11, 14–16, 19–22

## L04 Potential and exact intensity budget

**DERIVED IDENTITY.** The pre-stage increment is  $-\nabla\mathcal{V}$ , with  $\mathcal{V} = \varepsilon \sum_i (s_i^2/4 - k_i s_i/2) + (g/2) \sum_{\{i,j\}} |z_i - z_j|^2$ ,  $s_i = |z_i|^2$ . The complete intensity increment includes the squared norm of the pre-increment.

**Domain:** Triad; real Cartesian gradient; exact normalization.

**Authority/proof:** GR0 §4(d)–(e); main text §2.

**Prior ownership:** Yes: Papers B/D/E; Atlas 07

**Qualification:** No universal discrete descent: common input 2 at  $\text{eps}=k=1, g=0$  maps to -4.

**Interpretation limit:** Not a calibrated physical energy or elastic shell energy.

**Supporting predicates:** GR0 ordinals 23–29, 31–32

## L05 Positive synchronized fixed circle

**EXACT THEOREM.** For  $\varepsilon \neq 0$ , fixed states on the nonzero common-channel line form  $\sqrt{k}e^{ix}\mathbf{1}$  exactly when  $k > 0$ .

**Domain:** Any admitted finite ring; real  $g, \text{lambda}$ .

**Authority/proof:** GR0 §5.1; main text §3.

**Prior ownership:** Yes in unit/triad cases; general equal-k formulation consolidated by GR0

**Qualification:** At  $\text{epsilon}=0$  every common amplitude is fixed and an extra neutral direction appears.

**Interpretation limit:** A fixed circle in state space is not a material orbit.

**Supporting predicates:** Analytic proof/source statement; no individually matched predicate claimed.

## L06 Complete homogeneous derivative

**DERIVED IDENTITY.** The amplitude block is  $(1 - 2\varepsilon k)I + g\Delta$ ; the phase block is  $(I + 3\lambda\Delta)(I + g\Delta)$ .

**Domain:** Positive equal-k background; local real amplitude/phase chart.

**Authority/proof:** GR0 §5.2; main text §3.

**Prior ownership:** Unit triad known in Paper F; all-ring equal-k derivation GR0

**Qualification:** Stage order matters away from homogeneous commutation.

**Interpretation limit:** No inertia is added.

**Supporting predicates:** GR0 ordinals 33–35, 45

## L07 Finite Fourier spectrum

**EXACT THEOREM.**  $\mu_a = 1 - 2\epsilon k - gh_m$ ,  $\mu_\phi = (1 - gh_m)(1 - 3\lambda h_m)$ , and  $h_m = 4 \sin^2(\pi m/N)$ .

**Domain:**  $N=3q$ ,  $m=0, \dots, N-1$ , equal positive  $k$ .

**Authority/proof:** GR0 §5.3; main text §3.

**Prior ownership:** Cycle Fourier calculus and unit triad known; present synthesis from GR0

**Qualification:** Triad pullback modes have  $h=0, 3, 3$  and are not long-wave modes.

**Interpretation limit:** Real/even multipliers exclude propagating linear wave dispersion only on this branch.

**Supporting predicates:** GR0 ordinals 36–41

## L08 Normal stability

**EXACT THEOREM.** Strict normal contraction is equivalent to  $|\mu_a(m)| < 1$  for every  $m$  and  $|\mu_\phi(m)| < 1$  for nonzero  $m$ ; it yields local finite- $N$  attraction modulo common phase.

**Domain:** Fixed finite  $N$ ; nonzero chart; strict inequalities.

**Authority/proof:** GR0 §5.4; main text §3.

**Prior ownership:** Prior local methods in Paper F; scope GR0

**Qualification:** No uniform-in- $N$  basin; equality at unit modulus is undecided by the linear test.

**Interpretation limit:** Relaxation may include sign alternation.

**Supporting predicates:** GR0 ordinals 36–41

## L09 Restricted modal limit

**EXACT THEOREM.** For fixed  $m$  and  $D_\phi = g + 3\lambda > 0$ ,  $\mu_\phi(2\pi m/N)^{\lfloor \tau N^2 \rfloor}$  tends uniformly on bounded nonnegative  $\tau$  intervals to  $e^{-4\pi^2 D_\phi m^2 \tau}$ , with  $O(N^{-2})$  error.

**Domain:** Fixed native coefficients and fixed finite collection of Fourier indices;  $N \rightarrow \infty$  through  $3q$ .

**Authority/proof:** GR0 §6.1–6.2; main text §3.

**Prior ownership:** Not asserted as prior result; GR0

**Qualification:** No nonlinear continuum theorem or control of all growing modes;  $D_\phi=0$  has quartic leading decrement.

**Interpretation limit:** Diffusion-like modal analogy, no physical  $dt$  or length.

**Supporting predicates:** GR0 ordinals 42–44

## L10 Positive unequal branch and screening

**EXACT THEOREM.** The local analytic positive solution of  $\epsilon r_i(k_i - r_i^2) + g(\Delta r)_i = 0$  has  $(2\epsilon k I - g\Delta)u = \epsilon\sqrt{k}\kappa$ . For three-periodic zero-mean  $\kappa$ ,  $u = \epsilon\sqrt{k}\kappa/(2\epsilon k + 3g)$ .

**Domain:**  $k > 0$ ; strict finite- $N$  normal stability at equal point;  $k_i = k + \epsilon \kappa_i$ .

**Authority/proof:** GR1 §2, (2)–(6); main text §4.

**Prior ownership:** Fixed equation known Paper A §7.5; response formula GR1

**Qualification:** Positive inverse derivative, local uniqueness only; no global branch continuation.

**Interpretation limit:** Screened-Poisson structural analogy only when  $g > 0$ .

**Supporting predicates:** GR1 ordinals 1, 5–6

## L11 Pre-stage weighted response

**EXACT THEOREM.**  $P = R^{-1}H_r R$ ,  $H_r r = r$ , and  $R^2 P = P^T R^2$ . The inner product has weights  $r_i^2$  and Dirichlet form  $g \sum_{\{i,j\}} r_i r_j (f_i - f_j)^2$ .

**Domain:** Every positive fixed branch; real  $g$ ; uniqueness of weights up to scale when  $g$  nonzero and graph connected.

**Authority/proof:** GR1 §3, (7)–(12); main text §5.

**Prior ownership:** Derivative framework prior; explicit weighting/classification GR1

**Qualification:**  $g=0$  permits every positive diagonal weight;  $g < 0$  reverses form sign; stochasticity needs entrywise nonnegativity.

**Interpretation limit:** Background-dependent mathematical response geometry, not spacetime.

**Supporting predicates:** GR1 ordinals 1–4

## L12 Full unequal derivative

**DERIVED IDENTITY.**  $J_\phi = S_\phi P$ , with  $S_\phi = I + \ell \Delta$ ,  $\ell = 3\lambda$ ; the amplitude block is  $H_r - 2\epsilon R^2$ . There is no first-order amplitude/phase mixing.

**Domain:** Positive branch; phase coordinates  $b = R \phi$ .

**Authority/proof:** GR1 §3, (13)–(15); main text §5.

**Prior ownership:** Prior derivative framework; GR1 full formulation

**Qualification:** Cartesian imaginary block is  $R S_\phi R^{-1} H$ ; not SP in those coordinates.

**Interpretation limit:** Two separately self-adjoint factors need not have a common metric.

**Supporting predicates:** GR1 ordinals 7–12

### L13 Triad cycle factorization

**DERIVED IDENTITY.**  $C_{\text{cyc}} = g\ell(1 - 3\ell)(a - b)(a - c)(b - c)\mathcal{K}/(a^2b^2c^2)$ , where  $\mathcal{K} = g^2\ell\sigma_1^3 + [g^2(1 - 2\ell) - g\ell]\sigma_1\sigma_2 + (3g\ell - g + \ell)\sigma_3$ .

**Domain:**  $a, b, c > 0$ ; exact algebra for any real  $g, \ell$ .

**Authority/proof:** GR1 §4, (16)–(20); main text §6.

**Prior ownership:** GR1; no historical priority claim

**Qualification:** GR2 (36) omits a '+' in its restatement; this paper uses verified GR1 (20).

**Interpretation limit:** Algebraic response circulation, not physical current.

**Supporting predicates:** GR1 ordinals 13

### L14 Triad diagonal criterion and exceptions

**EXACT THEOREM.** With positive off-diagonals,  $C_{\text{cyc}} = 0$  iff positive diagonal balance exists. One choice is  $W = \text{diag}(J_{21}J_{31}, J_{12}J_{31}, J_{13}J_{21})$ . Exceptional factors are  $g = 0$ ,  $\ell = 0$ ,  $\ell = 1/3$ , equal radius pairs, or  $\mathcal{K} = 0$ .

**Domain:** Sufficiently near a strictly stable equal triad, guaranteeing positive off-diagonals.

**Authority/proof:** GR1 §4, (18)–(24); main text §6.

**Prior ownership:** GR1 classification; cycle logic standard and reproved

**Qualification:** Outside positive-entry neighbourhood zero cycle alone does not prove positive weights.

**Interpretation limit:** Locality of weights is a mathematical restriction.

**Supporting predicates:** GR1 ordinals 14–27

### L15 Cubic onset

**DERIVED IDENTITY.**  $C_{\text{cyc}} = g\ell(1 - 3\ell)K_0[\varepsilon/(2\varepsilon k + 3g)]^3 V(\kappa)\eta^3 + O(\eta^4)$ , with  $K_0 = g(9g - 1) + \ell(1 - 3g)^2$ .

**Domain:** Weak branch of result 10; three-periodic zero-mean perturbation.

**Authority/proof:** GR1 §5, (25)–(26); main text §6.

**Prior ownership:** GR1

**Qualification:** If  $K_0 = 0$  the obstruction is  $O(\eta^5)$  or smaller, not automatically zero.

**Interpretation limit:** First-order fitted weights cannot settle exact diagonal balance.

**Supporting predicates:** GR1 ordinals 13, 15

### L16 Larger-ring diagonal impossibility

**EXACT THEOREM.** For  $N \geq 6$  and  $g\ell \neq 0$ , distinct positive repeated radii forbid positive diagonal balance. For  $(a, a, c)$ , it exists iff  $\ell - g - g\ell(2a/c + c/a) = 0$ , and then  $W = R$ .

**Domain:**  $N = 3q \geq 6$ ; positive three-periodic branch; no entrywise positivity/stability assumption.

**Authority/proof:** GR1 §6, (27)–(30); main text §6.

**Prior ownership:** GR1

**Qualification:**  $g=0, \ell=0$ , or equal radii remain reversible; triad exceptional loci do not suffice on the full ring.

**Interpretation limit:** Distance-two response links strengthen the node-local obstruction.

**Supporting predicates:** GR1 ordinals 28–38

### L17 SPD criterion

**EXACT THEOREM.** A real  $J$  admits SPD  $G$  with  $GJ = J^T G$  iff its spectrum is real and semisimple. All solutions are  $G = V^{-T}KV^{-1}$ , with  $K$  SPD and commuting with the real eigenvalue matrix. Positive sums of  $E_\rho^T E_\rho$  also construct solutions.

**Domain:** Any finite real matrix; repeated semisimple/zero roots allowed.

**Authority/proof:** GR2 §2, (4)–(6); main text §7.

**Prior ownership:** Standard finite-dimensional lemma, reproved; not a novelty claim

**Qualification:** Defective real matrices fail as well as complex-spectrum matrices.

**Interpretation limit:**  $G$  is nonunique and need not be node-local or physically privileged.

**Supporting predicates:** Analytic proof/source statement; no individually matched predicate claimed.

### L18 Exact triad reality

**EXACT THEOREM.**  $M_3 = \sigma R^{-1} - \mathbf{1}\mathbf{1}^T$  is symmetric positive semidefinite with kernel  $r$ ; its positive roots have sum  $h$  and product  $t$ . The quotient polynomial is  $Q(\xi) = \xi^2 - d(2 - gh)\xi + d^2(1 - gh + g^2t)$ , with  $D_3 = d^2g^2(h^2 - 4t) \geq 0$ .

**Domain:** Positive  $a, b, c$ ; arbitrary real  $g, \ell$ ;  $d = 1 - 3\ell$ ; transverse means quotient by common phase.

**Authority/proof:** GR2 §3, (7)–(13); main text §8.

**Prior ownership:** GR2

**Qualification:**  $h^2 - 4t = 0$  iff radii equal;  $g=0$  or  $d=0$  also gives semisimple transverse coincidence.

**Interpretation limit:** Triad diagonal obstruction does not force complex roots.

**Supporting predicates:** GR2 ordinals 1–5

## L19 Full triad resonance defect

**EXACT THEOREM.** For  $M_3 w_\alpha = \mu_\alpha w_\alpha$ ,  $w_\alpha \perp r$ , set  $v_\alpha = R^{-1} w_\alpha$ ,  $p_\alpha = 1 - g\mu_\alpha$ . A full defect occurs at  $dp_\alpha = 1$  with  $\ell p_\alpha \mathbf{1}^T v_\alpha \neq 0$ .

**Domain:** Positive nonconstant background; degenerate g,ell,d cases treated separately.

**Authority/proof:** GR2 §3, (14)–(15); main text §8.

**Prior ownership:** GR2

**Qualification:** Three distinct radii imply a defective resonance; equal-pair antisymmetric collision is semisimple. Local strict stability excludes eigenvalue-one resonance.

**Interpretation limit:** Transverse quotient semisimplicity is not always full-matrix semisimplicity.

**Supporting predicates:** GR2 ordinals 17–18

## L20 Triad discriminant onset

**DERIVED IDENTITY.**  $\mathcal{D}_3 = 6d^2 g^2 [\varepsilon/(2\varepsilon k + 3g)]^2 \sum_i \kappa_i^2 \eta^2 + O(\eta^3)$ . No order produces complex transverse roots on a positive triad.

**Domain:** Weak branch; dg and kappa nonzero for positive leading term.

**Authority/proof:** GR2 §4, (16); main text §8.

**Prior ownership:** GR2

**Qualification:** dg=0 leaves exact real coincidence.

**Interpretation limit:** Independent from the cubic diagonal-cycle obstruction.

**Supporting predicates:** GR2 ordinals 12

## L21 Finite Bloch blocks

**DERIVED IDENTITY.** For  $\phi_{n,\alpha} = z^n v_\alpha$ ,  $z^q = 1$ , the exact block is  $J(z) = (I + \ell D(z)) R^{-1} H_r(z) R$ , where the corner entries of the cycle block  $D(z)$  are  $z^{-1}$ ,  $z$ , its other off-diagonals are one and its diagonal is minus two.

**Domain:** N=3q; repeated a,b,c>0; exact finite Fourier transform under translation by three.

**Authority/proof:** GR2 §5, (17); main text §9.

**Prior ownership:** GR2 specialization; finite Fourier decomposition standard

**Qualification:** Conjugate blocks pair; each block need not have real coefficients.

**Interpretation limit:** No continuum approximation.

**Supporting predicates:** GR2 ordinals 19, 21, 23–24, 26, 28, 30, 32

## L22 Bloch polynomial and circulation coefficient

**DERIVED IDENTITY.**  $\det(\xi I - J(z)) = \xi^3 - T_B \xi^2 + B_B(z) \xi - D_B(z)$ , with coefficients in (9.7)–(9.9). In particular  $\Gamma_B = \ell g[\ell(1 + 3g) - g](a - b)(a - c)(b - c)/(abc)$  and  $\text{Im } B_B = -\Gamma_B \sin \theta$ .

**Domain:** Same finite positive branch; all real parameters;  $|z|=1$ .

**Authority/proof:** GR2 §5, (18)–(20); main text §9.

**Prior ownership:** GR2

**Qualification:** Gamma\_B differs from the triad cycle factor and the chiral scalar Gamma\_ch.

**Interpretation limit:** Spectral response coefficient, not physical flux.

**Supporting predicates:** GR2 ordinals 6–11

## L23 Exact block decision

**EXACT THEOREM.** Nonzero  $\Gamma_B \sin \theta$  forces non-real roots. For real cubic coefficients, positive discriminant gives three simple real roots, negative gives a complex pair, and zero requires a rank/minimal-polynomial test.

**Domain:** Every admitted Bloch block; full ring SPD iff all blocks have real semisimple spectra.

**Authority/proof:** GR2 §5, (21)–(23); main text §9.

**Prior ownership:** GR2 application; cubic/rank tests reproved

**Qualification:** At a double root rho with distinct sigma,  $\text{rank}(J - \rho I) = 1$  is required; at triple root  $J = \rho I$  is required.

**Interpretation limit:** Non-real does not imply defective or unstable.

**Supporting predicates:** GR2 ordinals 20, 22, 25, 27, 29, 31

## L24 Generic larger-ring complex response

**EXACT THEOREM.** For  $q \geq 3$ , three distinct radii and  $g\ell[\ell(1 + 3g) - g] \neq 0$  force non-real spectrum.  $\Gamma_B$  is cubic in weak inequality; simple nonzero roots have the cubic imaginary shift in Section 9.

**Domain:** N>=9; repeated weak branch; simple-root expansion needs distinct nonzero equal-background root.

**Authority/proof:** GR2 §5, (24)–(27); main text §9.

**Prior ownership:** GR2

**Qualification:** Vanishing Gamma\_B is not sufficient for real semisimple spectrum; stability can persist.

**Interpretation limit:** Oscillatory linear response only.

**Supporting predicates:** GR2 ordinals 9, 27, 31

## L25 Generic weak six-ring reality

**DERIVED IDENTITY.** At  $z = -1$ ,  $\mathfrak{D}_\pi = 6g^2(1 - \ell)^2 \sum_i v_i^2 (3E_\pi)^4 \eta^2 + O(\eta^3)$ , where  $E_\pi = g + \ell - 5g\ell$ .

**Domain:** Fixed stable equal N=6 baseline; g(1-ell)Epi nonzero and nontrivial inequality.

**Authority/proof:** GR2 §6, (29)–(30); main text §10.

**Prior ownership:** GR2

**Qualification:** Neighbourhood not uniform near exceptional baselines. At Epi=0 a separate first-order test applies; ell=1 needs rank test.

**Interpretation limit:** Six and twelve rings need not share spectral reality.

**Supporting predicates:** GR2 ordinals 13–16

## L26 Exact six-ring nilpotent fixture

**EXACT THEOREM.** For  $r = (1 - \tau, 1, 1 + \tau)$ ,  $g = (1 - \tau^2)/4$ ,  $\ell = 1$ , the exact block is  $J(-1) = (\tau^2/4)(1, -1, 1)^T(1 + \tau, 2, 1 - \tau)$ . It has rank one, square zero, and Jordan sizes 2 + 1.

**Domain:**  $0 < |\tau|$  sufficiently small; eps=1/20; k\_i from exact native fixed equations.

**Authority/proof:** GR2 §6, (31)–(33); main text §10.

**Prior ownership:** GR2

**Qualification:** At tau=1/1000 the exact contraction box links fixture to weak native branch; varying tau does not hold kappa fixed.

**Interpretation limit:** A normally stable response can be defective.

**Supporting predicates:** GR2 ordinals 33–34, 36, 41–42

## L27 Six-ring complex unfolding

**EXACT THEOREM.** With the same  $r, g, k_i, \varepsilon$  and  $\ell = 1 + \delta$ , the coefficients are (10.7), and  $\mathfrak{D}_\pi = (27/2)\tau^{12}\delta^3 + O(\delta^4)$ .

**Domain:** Fixed nonzero small tau; sufficiently small delta.

**Authority/proof:** GR2 §6, (34)–(35); main text §10.

**Prior ownership:** GR2

**Qualification:** delta<0 complex pair; delta=0 defect; delta>0 distinct real roots; exact rational signs checked at  $\pm 10^{-14}$  for tau=10<sup>-3</sup>.

**Interpretation limit:** Modes can remain inside unit disk.

**Supporting predicates:** GR2 ordinals 35, 37–38

## L28 Actual spatial representation

**DEFINITION / ADOPTION.** The generators act by  $C_3 : (q, p) \mapsto (Pq, Pp)$ ,  $\sigma_v : (q, p) \mapsto (-Q_v q, Q_v p)$ , and  $\sigma_h : (q, p) \mapsto (q, -p)$ .

**Domain:** Fixed accepted shell frames; spatial polar/axial laws distinct.

**Authority/proof:** CM0 §2, actual mirror-action table; main text §11.

**Prior ownership:** Yes: Paper B conventions, reused by CM0

**Qualification:** Not ordinary S3 action on both channel triples. This is kinematic covariance; F carries k under permutations and has a regular-domain qualification for mirrors involving common phase.

**Interpretation limit:** Kinematic symmetry does not silently give full fixed-unequal-k dynamical symmetry.

**Supporting predicates:** CM0 ordinals 1–39

## L29 Ambient polynomial multiplicities

**EXACT THEOREM.** Real-linear polar and axial map dimensions are 2 and 2; homogeneous quadratic dimensions are 5 and 3. Complete bases appear in Theorem 11.1.

**Domain:** Real polynomials in six state coordinates, full D3h, fixed geometry.

**Authority/proof:** CM0 §3; §4 for sectors; main text §11.

**Prior ownership:** CM0 census; no priority claim

**Qualification:** C3 alone admits many more maps; mixtures' images are not sums of separate images.

**Interpretation limit:** No unique core observable follows.

**Supporting predicates:** CM0 ordinals 4–44

### L30 Optional common-phase invariance

**EXACT THEOREM.** The optional common-phase-invariant quadratic polar space is spanned by  $N_f(au+bv)$ ,  $N_f(u^2+v^2)$ ,  $\Gamma_{\text{ch}}e_z$ . The axial invariant space is spanned by  $W$ .

**Domain:** Additional common-phase invariance imposed as a kinematic requirement.

**Authority/proof:** CM0 §3, (8)–(9); main text §11.

**Prior ownership:** CM0 classification;  $W$  definition already accepted

**Qualification:** There is no nonzero invariant real-linear map; this is not a globally valid phase quotient of  $F$  at zeros.

**Interpretation limit:** Even extra phase invariance does not select a polar position observable.

**Supporting predicates:** CM0 ordinals 45

### L31 Independent chiral projections

**EXACT THEOREM.**  $C_{\text{ch}} = q \times p = e \times (av - bu) + u \times v$ . Its horizontal-axial projection is  $W = T_f C_{\text{ch}} = \sqrt{3}(bN_f u - aN_f v)$ ; its vertical-polar projection is  $(e^T C_{\text{ch}})e_z = \Gamma_{\text{ch}}e_z$ .

**Domain:** D3h shell representation; degree-two antisymmetric bilinears.

**Authority/proof:** CM0 §4, (13); SA0 §6; main text §11.

**Prior ownership:**  $C$  and  $W$  accepted earlier; complete two-projection statement CM0

**Qualification:** Neither projection alone determines  $C$ ; raw  $C$  is not ambient; axial parity need not imply chirality.

**Interpretation limit:** No physical rotation axis.

**Supporting predicates:** CM0 ordinals 46–47

### L32 Observable kernels and images

**DERIVED IDENTITY.** For transverse  $u, v$ ,  $\|N_f(u^2)\|^2 = \|u\|^4/4$ ,  $\|N_f(uv)\|^2 = \|u\|^2\|v\|^2/4$ , and  $\Gamma_{\text{ch}}^2 = 3(\|u\|^2\|v\|^2 - (u \cdot v)^2)$ . Products inside  $N_f$  are entrywise.

**Domain:**  $u, v$  perpendicular to  $e$ ; native fixed  $N$ .

**Authority/proof:** CM0 §4, (10)–(12) and generator tables; main text §11.

**Prior ownership:** CM0

**Qualification:** Nonlinear kernels are zero sets; mixture images must be computed jointly.

**Interpretation limit:** Nonzero spatial readout at a fixed state is not displacement per step.

**Supporting predicates:** CM0 ordinals 48–50

### L33 Observable increments and response

**DERIVED IDENTITY.**  $\Delta V = V(F(\Omega)) - V(\Omega)$ . Polynomial increments follow substitution; local response at a fixed background is  $DV(\Omega_*)DF(\Omega_*)$ .

**Domain:** Degree  $\leq 2$  basis; equal or weak unequal positive branch for linear response.

**Authority/proof:** CM0 §5, (14)–(20) and response tables; main text §11.

**Prior ownership:** CM0 application of chain rule

**Qualification:** Generic basis readouts can be generated from zero; not preserved or autonomous in general.

**Interpretation limit:** Delta  $V$  is not velocity or acceleration.

**Supporting predicates:** CM0 ordinals 53–78

### L34 Invertible shell attachment

**EXACT THEOREM.**  $D : \mathbb{C}^3 \rightarrow \bigoplus_i W_i$  has inverse  $E$ . The full section evolves by  $D F E$ , with transported graph increment and outward-normal phase rotation.

**Domain:** Exactly three labelled tangent fibres; fixed centres and frames.

**Authority/proof:** SA0 §§2–3; §8.1; main text §12.

**Prior ownership:** Yes: Paper B; SA0 types the inherited structure

**Qualification:** No continuous polygon field, deformation, seam/rim data or physical position follows.

**Interpretation limit:** Only the discrete section exists as a native state representation.

**Supporting predicates:** SA0 ordinals 176–211

### L35 Two distinct seam criteria

**EXACT THEOREM.** For constant face extensions, ambient equality at every seam requires  $q = 0, p = be$ ; matched parallelism requires  $q = ae, p = be$ .

**Domain:** Conditional extension criteria on the canonical three vertical planes.

**Authority/proof:** SA0 §3.3; main text §12.

**Prior ownership:** SA0 comparison, not new native seam law

**Qualification:** Neither condition is imposed; not a classification of every intrinsic notion of continuity.

**Interpretation limit:** No physical seam law is inferred.

**Supporting predicates:** SA0 ordinals 212–216

### L36 Finite-carrier census

**EXACT THEOREM.** Degree (0,1,2) counts are face scalar/normal (1,1,8), tangent (0,4,8), ambient-face (1,5,16), vertical seam (0,2,4), rim scalar (1,1,5), and rim vector (1,3,12).

**Domain:** Full D3h, polynomial state maps, finite carriers; seam labels via opposite face; whole-rim values.

**Authority/proof:** SA0 §5.1–5.6; main text §12.

**Prior ownership:** SA0; centre counts reused CM0

**Qualification:** Pointwise fields require a spatial function space; values on loops need not be tangent everywhere.

**Interpretation limit:** Multiplicity cannot supply physical selection.

**Supporting predicates:** SA0 ordinals 154–173

### L37 Preparation type separation

**DERIVED IDENTITY.** Lens and incident data pass through  $\mathbb{C}^2 \otimes \mathbb{C}^3$ , fixed  $U^n$ , extraction  $\chi^\dagger \otimes I$ ,  $\Omega_0$ , then  $D\Omega_0$ . Preparation internals have no material shell attachment and no feedback enters F.

**Domain:** Named existing response, tensor order, branch and initialization receipt.

**Authority/proof:** SA0 §7.1–7.2; main text §12.

**Prior ownership:** Preparation formula prior Atlas 08; unified separation SA0

**Qualification:** Final Omega does have the existing D attachment;  $\mathbb{C}^6$  has twelve real dimensions.

**Interpretation limit:** Initialization is not boundary dynamics.

**Supporting predicates:** SA0 ordinals 223–226

### L38 Fibre closure criterion

**EXACT THEOREM.**  $\mathcal{A}F = F_{\mathcal{A}}\mathcal{A}$  on the image exists iff  $\mathcal{A}(x) = \mathcal{A}(x')$  implies  $\mathcal{A}(Fx) = \mathcal{A}(Fx')$ .

**Domain:** Any set maps; restrict domain explicitly for local/regular versions.

**Authority/proof:** SA0 §8.1; main text §13.

**Prior ownership:** Elementary quotient criterion reproved, no priority claim

**Qualification:** No smoothness, locality or invertibility conclusion.

**Interpretation limit:** Closure is not physical observability.

**Supporting predicates:** Analytic proof/source statement; no individually matched predicate claimed.

### L39 Nonclosed lower readouts

**EXACT THEOREM.** Single-face values, intensities, raw chirality, W, Gamma and joint (W,Gamma) generally fail closure. At  $g = 0$ ,  $I_i^+ = I_i[1 + \varepsilon(k_i - I_i)]^2$ .

**Domain:** Exact native witnesses in SA0; parameters fixed in each comparison.

**Authority/proof:** SA0 §8.1–8.2; main text §13.

**Prior ownership:** CM0 had ambient witnesses; these explicit fibre comparisons SA0

**Qualification:** Do not claim every combination/parameter locus is nonclosed.

**Interpretation limit:** Lost state information obstructs an autonomous readout law.

**Supporting predicates:** SA0 ordinals 219–222

### L40 Complex coherence quotient

**EXACT THEOREM.**  $\mathcal{H} = \Omega\Omega^\dagger$  closes when all diagonal entries of  $M\mathcal{H}M^T$  are positive:  $\mathcal{H}^+ = U_\delta M\mathcal{H}M^T U_\delta^\dagger$ , with delta from imaginary cubes of normalized transposed coherence entries.

**Domain:** Rank $\leq 1$  PSD Hermitian H; real native parameters; regular pre-sync domain.

**Authority/proof:** SA0 §8.3, first theorem; main text §13.

**Prior ownership:** SA0 synthesis result

**Qualification:** One-step domain need not be invariant; zero-stratum obstruction retained.

**Interpretation limit:** Channel coherence is not a spatial metric.

**Supporting predicates:** Analytic proof/source statement; no individually matched predicate claimed.

### L41 Real Gram quotient

**EXACT THEOREM.**  $S_G = qq^T + pp^T$  also closes on the regular pre-stage domain: equal Gram factors differ by  $O(2)$ , namely common phase and/or conjugation.

**Domain:** Rank $\leq 2$  real PSD Gram matrix; same regularity test.

**Authority/proof:** SA0 §8.3, second theorem; main text §13.

**Prior ownership:** SA0

**Qualification:**  $S\_G$  loses chiral orientation; autonomy follows from symmetry, not injectivity.

**Interpretation limit:** A nonlocal channel quotient, not one scalar per face.

**Supporting predicates:** Analytic proof/source statement; no individually matched predicate claimed.

## L42 Global zero-stratum failure

**EXACT THEOREM.** For  $\varepsilon = g = 0$ ,  $\lambda = 1/10$ , inputs  $(1, -1, 0)$  and  $e^{i\pi/6}(1, -1, 0)$  have identical coherences and real Grams. Outputs have  $\mathcal{H}_{12}^+ = -1$  versus  $-e^{-i/5}$ , and  $S_{G,12}^+ = -1$  versus  $-\cos(1/5)$ .

**Domain:** Native Arg0 convention; same exact parameter tuple.

**Authority/proof:** SA0 §8.4; main text §13.

**Prior ownership:** SA0 explicit closure witness; zero convention caveat prior Paper A/B

**Qualification:** At  $\lambda=0$  both Gram congruences close globally.

**Interpretation limit:** No repaired zero convention is introduced.

**Supporting predicates:** Analytic proof/source statement; no individually matched predicate claimed.

## L43 Retained computational evidence

**NUMERICAL EVIDENCE.** Counts are GR0 45, GR1 38, GR2 42, CM0 78, SA0 226, total 429 heterogeneous exact/source checks. Native finite differences, 100-digit evaluations and residuals remain separate datasets.

**Domain:** Only bounded fixtures and precisions actually recorded.

**Authority/proof:** ALL GR0 §7; GR1 §7; GR2 §8; CM0 §7; SA0 §9; main text §17.

**Prior ownership:** Retained evidence, not newly rerun or new theorems

**Qualification:** Predicates are not proofs; small native residuals cannot resolve every tiny imaginary part.

**Interpretation limit:** No broad numerical survey or certification is claimed.

**Supporting predicates:** Analytic proof/source statement; no individually matched predicate claimed.

## L44 Interpretation boundary

**PHYSICAL INTERPRETATION.** Response inner products, screening, complex modes, circulation, vector parity, face sections and initialization do not by themselves define the corresponding physical metric, gravity, wave, current, position, deformation or flux.

**Domain:** Claims confined to the finite native mathematical model.

**Authority/proof:** ALL Checkpoint conclusions and explicit limits; main text §14.

**Prior ownership:** Consistent with accepted paper boundaries

**Qualification:** Absence in this model is not a universal no-go theorem.

**Interpretation limit:** No new physical theory is established.

**Supporting predicates:** Analytic proof/source statement; no individually matched predicate claimed.

## L45 Open interfaces

**OPEN QUESTION.** Global branch/zero-stratum analysis, nonlinear uniform limits, privileged observable/metric selection and minimal sufficient coherence subsets remain open.

**Domain:** No new research undertaken for this publication synthesis.

**Authority/proof:** ALL GR2 §9; CM0 §6; SA0 §10 question 6; main text §15.

**Prior ownership:** Open, not claimed results

**Qualification:** Existing finite proofs do not solve these interfaces.

**Interpretation limit:** Research publication stops at synthesis.

**Supporting predicates:** Analytic proof/source statement; no individually matched predicate claimed.

## C Triad algebra

The principal minors of  $M_3 = \sigma R^{-1} - ee^T$  sum to

$$\sum_{i < j} \left[ \left( \frac{\sigma}{r_i} - 1 \right) \left( \frac{\sigma}{r_j} - 1 \right) - 1 \right] = \frac{\sigma(a^2 + b^2 + c^2)}{abc} = t.$$

Its trace is  $h$ , and its zero eigenvalue is supplied by  $M_3 r = 0$ . This derives (8.4) without treating the sum-zero plane as invariant for unequal  $J$ . For  $(a, a, c)$ , the antisymmetric equal-pair mode has  $\mu_A = 2 + c/a$ , while the other transverse mode has  $\mu_B = c/a + 2a/c$ ; substituting in (8.7) decides full-matrix collisions.

The alternating cycle expression and the imaginary Bloch coefficient are different invariants:

$$C_{\text{cyc}} = \frac{g \ell d V_r}{a^2 b^2 c^2} \mathcal{K}, \quad \Gamma_B = \frac{\ell g [\ell(1 + 3g) - g] V_r}{abc}.$$

In particular their exceptional parameter factors are not interchangeable. This paper uses the plus sign before  $(3g\ell - g + \ell)\sigma_3$  in verified GR1 (20); the missing plus in the later GR2 (36) restatement is recorded as a transcription correction, not a new result or a modification of GR2.

## D Bloch coefficient derivation

Set  $z \neq 0$  formally before specializing to roots of unity. Compute  $J$  from (9.3), then

$$T_B = \sum_i J_{ii}, \quad B_B = \sum_{i < j} (J_{ii}J_{jj} - J_{ij}J_{ji}), \quad D_B = \det J.$$

The first contains no  $z$ . The second is Laurent-linear in  $z$ :

$$B_B(z) = B_B(1) + C_B(z + z^{-1} - 2) - \frac{\Gamma_B}{2}(z - z^{-1}).$$

The antisymmetric coefficient is obtained by subtracting the expression at  $z^{-1}$ ; its numerator is alternating in a,b,c and factors as  $-\ell g[\ell(1 + 3g) - g]V_r(z - z^{-1})/(abc)$ . The symmetric coefficient reduces to the stated  $C_B$ . The determinant follows from the product of the two stage determinants, proving (9.7)–(9.9). These operations, rather than eigenvalue fitting, are reproduced by the paper-local checker.

At  $N=6$  the blocks are  $1, -1$ ; at  $N=12$  they are  $1, i, -1, -i$ . Exact Fourier injection matrices have columns  $z^n e_\alpha$ ; direct multiplication  $J_N B_z = B_z J(z)$  verifies the decomposition. The characteristic polynomial of the full matrix is the product of block polynomials, as this invariant direct sum spans every coordinate.

## E Exact six-ring fixture

At  $\tau = 1/1000$ ,  $\varepsilon = 1/20$ ,  $g = 999999/4000000$ , the native coefficients in (10.2) are

$$(k_1, k_2, k_3) = (491493/500000, 1, 508493/500000), \quad k = 749993/750000.$$

They lie strictly between  $(49/50)^2$  and  $(51/50)^2$ . Let  $L = 49/50, U = 51/50$ . On the real box  $r_i \in [L, U]$ , the amplitude map has off-diagonal derivatives  $g$  and diagonal derivatives  $1 + \varepsilon(k_i - 3r_i^2) - 2g$ , all nonnegative for this fixture and the interpolation of coefficients from their mean. Its row sum is  $1 + \varepsilon(k_i - 3r_i^2) \leq 9067893/10^7 < 1$ . Monotonicity reduces box invariance to the constant endpoints: their coupling vanishes, the lower endpoint increments are nonnegative because  $k_i > L^2$ , and upper endpoint increments are nonpositive because  $k_i < U^2$ . This proves contraction and existence/uniqueness in the box throughout the interpolation, connecting the exact fixture to the local positive equal branch.

At the defect, the row-column pairing in (10.3) is  $(1 + \tau) - 2 + (1 - \tau) = 0$ . Since the matrix is nonzero, its minimal polynomial is  $\xi^2$  and characteristic polynomial  $\xi^3$ . Exact rational substitution of  $\delta = \pm 10^{-14}$  into (10.7) gives discriminants with the respective signs. Their full rational values are retained in GR2 and the package evidence; the sign is not inferred from rounded eigenvalues.

## F Representation decomposition and observable kernels

For an orthogonal finite-group representation  $X$  and target  $Y$ , averaging its action on  $\text{Hom}(\text{Sym}^d X, Y)$  projects onto equivariants. Its trace is the dimension:

$$\frac{1}{12} \sum_{g \in D_{3h}} \chi_{\text{Sym}^d X}(g) \chi_Y(g), \quad \chi_{\text{Sym}^2 X}(g) = \frac{\chi_X(g)^2 + \chi_X(g^2)}{2}.$$

This both proves the multiplicity computation and gives a finite exact verification route, with no numerical representation fitting.

For transverse  $u, v$ ,

$$\|N_f(u \odot u)\|^2 = \frac{1}{4} \|u\|^4, \quad \|N_f(u \odot v)\|^2 = \frac{1}{4} \|u\|^2 \|v\|^2, \\ \Gamma_{\text{ch}}^2 = 3(\|u\|^2 \|v\|^2 - (u \cdot v)^2).$$

These follow by  $u_3 = -u_1 - u_2, v_3 = -v_1 - v_2$  and the exact frame matrix. If  $N_f u = (X, Y, 0)$ , then

$$N_f(u \odot u) = \frac{1}{3}(2XY, X^2 - Y^2, 0);$$

polarization gives  $N_f(u \odot v) = \frac{1}{3}(XW + YZ, XZ - YW, 0)$  for  $N_f v = (Z, W, 0)$ . The first is onto the plane as a complex-square map; the second is invertible in  $v$  for fixed nonzero  $u$ . Thus its kernel is  $u=0$  or  $v=0$ . Gamma vanishes exactly for dependent transverse vectors. Mean-times-transverse maps vanish when either factor vanishes. Linear maps  $T_f u, be_z, T_f v, ae_z$  have the evident transverse/mean kernels. Images of mixtures cannot be obtained by independently adding images with shared inputs.

## G Complete finite-carrier census

Define  $Kx = e \times x/\sqrt{3}$ ,  $K^2 = -\Pi$ , and

$$\mathcal{H}_s = (a^2, b^2, \|u\|^2, \|v\|^2), \quad \mathcal{E}_s = (au, \Pi(u^2), bv, \Pi(v^2)), \quad \mathcal{O}_s = (av, bu, \Pi(uv)),$$

with entrywise products in this appendix.

**Face scalars and normal coefficients.** Bases at degrees zero, one and two are  $e$ ,  $Ku$ , and  $\{he : h \in \mathcal{H}_s\} \cup \mathcal{E}_s$ . Multiplying the  $i$ -th scalar by outward  $n_i$  gives the identical count for polar normal assignments.

**Face tangent vectors.** Represent outputs by coefficient pair  $(Q, P)$ . Linear basis is  $(ae, 0), (u, 0), (0, be), (0, v)$ . Quadratic basis is  $(Kf, 0)$ ,  $f \in \mathcal{E}_s$ , together with  $(0, \Gamma_{\text{ch}} e)$  and  $(0, Kh)$ ,  $h \in \mathcal{O}_s$ . There is no constant tangent map. Orthogonal normal/tangent decomposition gives counts 1, 5, 16 for ambient vectors per face.

**Seams.** Opposite-face indexing has the ordinary face permutation representation. Scalars and ambient vectors therefore have the corresponding face counts. Vertical seam coefficients transform like  $p$  and have linear basis  $be, v$  and quadratic basis  $\Gamma_{\text{ch}} e, K(av), K(bu), K\Pi(uv)$ .

**Whole rims.** Let  $f_\sigma = f_e + \sigma f_o$ ,  $\sigma = \pm 1$ . Even scalars have bases 1 and  $\mathcal{H}_s$ ; odd scalars have  $b$  and  $\Gamma_{\text{ch}}$ , at degrees one and two. For polar vectors  $V_\sigma = V_e + \sigma V_o$ ,  $V_e$  has the main polar basis, while  $V_o$  has constant  $e_z$ , linear  $N_f v$ , and quadratic basis  $\{T_f h : h \in \mathcal{O}_s\} \cup \{h e_z : h \in \mathcal{H}_s\}$ . The horizontal reflection exchanges rims; under a general element  $G e_z = \eta e_z$ , the odd target transforms as  $\eta G$ .

These lists are independent by mean/transverse and q-q/p-p/q-p separation;  $K$  is invertible on  $e^\perp$ . The character counts above exhaust their dimensions. They require no selected seam or boundary point, but they do not supply a pointwise material field.

## H Closure counterexamples and observable evolution

The rational comparisons in Section 13 supply exact fibre failures, not only changes along one trajectory. The zero state is fixed and every homogeneous polynomial observable vanishes there. CM0's bounded examples can therefore be used in the same way. For example, at  $\varepsilon = 1/20, g = 1/5, k = e, \lambda = 0$ ,

$$q = e, p = (1, -1, 0) \mapsto q^+ = (19/20, 19/20, 1), p^+ = (7/20, -7/20, 0).$$

Here  $T_f u$  and Gamma initially vanish, but the former is generated and  $\Gamma_{\text{ch}}^+ = 7/200$ . At

$$q = (1, -1, 0), p = (1, 1, -2),$$

one obtains  $q^+ = (7/20, -7/20, 0)$ ,  $p^+ = (7/20, 7/20, -1/2)$ . This generates the imaginary mean and  $W$  from zero, with  $W^+ = (-7/200, 7\sqrt{3}/200, 0)$ . Swapping  $q$  and  $p$  supplies the corresponding complementary witnesses; a real  $(1, 2, 3)$  maps to  $(8/5, 17/10, 6/5)$ , changing the direction of  $T_f q$ . The complete checkpoint list is retained; no trajectory scan is added.

For a homogeneous quadratic map  $V(x) = B(x, x)$ , polarization gives  $\Delta V = 2B(x, \Delta x) + B(\Delta x, \Delta x)$ . This is the exact common formula behind the individual increments. It supplies no independent observable law unless the fibre criterion is met.

## I Verification inventory

The complete measurements, including every stored high-precision digit and all exact rational fixtures, are retained in `retained_evidence.json`. The following tables round values for print only. Maxima below compare like quantities within a stated checkpoint, never across error types. The 429 retained exact/source predicates are enumerated separately; none is replaced by these residuals. Paper-local checks are an additional bounded algebra and reference audit, with their own machine-readable result.

### Native complex128 finite differences

For GR0 the error is the maximum entrywise error of the complete real Jacobian. For GR1 it is separated into phase and radial blocks; for GR2 only phase response was tested. These are different quantities.

| GR0 ring | Step                     | Complete Jacobian error   |
|----------|--------------------------|---------------------------|
| C3       | $1.00000 \times 10^{-4}$ | $5.00211 \times 10^{-10}$ |
| C3       | $1.00000 \times 10^{-5}$ | $1.33778 \times 10^{-11}$ |
| C3       | $1.00000 \times 10^{-6}$ | $6.12623 \times 10^{-11}$ |
| C12      | $1.00000 \times 10^{-4}$ | $5.02309 \times 10^{-10}$ |
| C12      | $1.00000 \times 10^{-5}$ | $2.87038 \times 10^{-11}$ |
| C12      | $1.00000 \times 10^{-6}$ | $1.62148 \times 10^{-10}$ |

| GR1 N | Step                     | Phase error               | Radial error              |
|-------|--------------------------|---------------------------|---------------------------|
| 3     | $1.00000 \times 10^{-4}$ | $1.62181 \times 10^{-10}$ | $5.00302 \times 10^{-10}$ |
| 3     | $1.00000 \times 10^{-5}$ | $1.62181 \times 10^{-12}$ | $7.91794 \times 10^{-12}$ |
| 3     | $1.00000 \times 10^{-6}$ | $1.62093 \times 10^{-14}$ | $1.63248 \times 10^{-10}$ |
| 12    | $1.00000 \times 10^{-4}$ | $1.61339 \times 10^{-10}$ | $5.00302 \times 10^{-10}$ |
| 12    | $1.00000 \times 10^{-5}$ | $1.61340 \times 10^{-12}$ | $5.35114 \times 10^{-12}$ |
| 12    | $1.00000 \times 10^{-6}$ | $1.61537 \times 10^{-14}$ | $1.63248 \times 10^{-10}$ |

| GR2 N | Step                     | Phase error               |
|-------|--------------------------|---------------------------|
| 3     | $1.00000 \times 10^{-6}$ | $1.62093 \times 10^{-14}$ |
| 6     | $1.00000 \times 10^{-6}$ | $1.61537 \times 10^{-14}$ |
| 12    | $1.00000 \times 10^{-6}$ | $1.61537 \times 10^{-14}$ |

GR0 uses  $\varepsilon = 1/20, g = 1/5, \lambda = 1/1000$ , with  $(k, \chi) = (1, 0)$  on C3 and  $(4, \pi/7)$  on C12. GR1/GR2 use the weak positive branch at  $k = 1, \kappa = (-1, 0, 1), \eta = .01$  and the same parameters. Native fixed residuals are zero in GR0, and at most  $1.11023 \times 10^{-16}$  in the two GR1 tests. GR0's native frame/transport comparison and potential comparison each give  $1.11023 \times 10^{-16}$ , as two separate quantities.

## 100-digit transcription finite differences

These checks use the mathematical transcription with mpmath, not a 100-digit execution of the complex128 native kernel.

| Method/domain | Step                      | Maximum error             |
|---------------|---------------------------|---------------------------|
| GR0 C3        | $1.00000 \times 10^{-10}$ | $5.00000 \times 10^{-22}$ |
| GR0 C3        | $1.00000 \times 10^{-20}$ | $5.00000 \times 10^{-42}$ |
| GR0 C12       | $1.00000 \times 10^{-10}$ | $5.00000 \times 10^{-22}$ |
| GR0 C12       | $1.00000 \times 10^{-20}$ | $5.00000 \times 10^{-42}$ |
| GR1 phase     | $1.00000 \times 10^{-10}$ | $1.62181 \times 10^{-22}$ |
| GR1 phase     | $1.00000 \times 10^{-20}$ | $1.62181 \times 10^{-42}$ |

## Weak-branch residuals and asymptotic coefficients

| $\eta$ | $\ r - r_0 - \eta u\ _\infty / \eta^2$ | $\mathcal{C}_{\text{cyc}} / \eta^3$ |
|--------|----------------------------------------|-------------------------------------|
| 0.01   | $2.00516 \times 10^{-2}$               | $-6.95490 \times 10^{-8}$           |
| 0.005  | $2.00477 \times 10^{-2}$               | $-6.95491 \times 10^{-8}$           |
| 0.0025 | $2.00457 \times 10^{-2}$               | $-6.95491 \times 10^{-8}$           |

GR1 fixed-equation residual has maximum  $1.00858 \times 10^{-99}$ .

GR1 cycle-factorization error has maximum  $1.11591 \times 10^{-103}$ .

GR1 pre-stage balance residual has maximum  $3.57184 \times 10^{-102}$ .

The proved leading cycle coefficient is  $-6.95491 \times 10^{-8}$ .

| $\eta$ | $\mathcal{D}_3/\eta^2$   | $\max  \operatorname{Im} \rho , z = i$ | $\max  \operatorname{Im} \rho /\eta^3$ |
|--------|--------------------------|----------------------------------------|----------------------------------------|
| 0.01   | $2.40509 \times 10^{-3}$ | $4.21737 \times 10^{-13}$              | $4.21737 \times 10^{-7}$               |
| 0.005  | $2.40510 \times 10^{-3}$ | $5.27172 \times 10^{-14}$              | $4.21738 \times 10^{-7}$               |
| 0.0025 | $2.40510 \times 10^{-3}$ | $6.58966 \times 10^{-15}$              | $4.21738 \times 10^{-7}$               |

GR2 fixed-equation residual has maximum  $1.01215 \times 10^{-99}$ .

GR2 characteristic residual has maximum  $1.82317 \times 10^{-101}$ .

The constructed triad SPD matrix has minimum eigenvalue  $9.98841 \times 10^{-1}$ , balance residual  $1.60733 \times 10^{-101}$ , and a simultaneous nonzero cycle obstruction  $-6.95490 \times 10^{-14}$ . Its complete entries and all block eigenvalues are retained separately. These 100-digit quantities resolve the small imaginary parts; the native finite-difference residuals alone would not certify them.

## Observable, quotient and preparation evidence

| CM0 observable derivative | Step                     | Maximum over 12 observables |
|---------------------------|--------------------------|-----------------------------|
| equal                     | $1.00000 \times 10^{-5}$ | $3.06800 \times 10^{-11}$   |
| weak unequal              | $1.00000 \times 10^{-5}$ | $1.12197 \times 10^{-11}$   |

The separate CM0 partial-zero chirality formula error is  $1.78351 \times 10^{-17}$ . The native decoder identity passed as a Boolean check, with no numerical residual invented.

| SA0 identity                        | Residual                  | Tolerance                 |
|-------------------------------------|---------------------------|---------------------------|
| Complex coherence quotient identity | $1.67250 \times 10^{-16}$ | $2.00000 \times 10^{-13}$ |
| real Gram closure under phase       | $5.55112 \times 10^{-17}$ | $2.00000 \times 10^{-13}$ |
| real Gram closure under conjugation | 0                         | $2.00000 \times 10^{-13}$ |
| zero-stratum witness formula        | $1.11022 \times 10^{-16}$ | $2.00000 \times 10^{-13}$ |
| canonical FaceState step            | 0                         | $1.00000 \times 10^{-15}$ |
| decoded view after step             | 0                         | $1.00000 \times 10^{-15}$ |
| SRG handoff direct product n=0      | 0                         | $2.00000 \times 10^{-13}$ |
| SRG handoff direct product n=1      | $1.09714 \times 10^{-17}$ | $2.00000 \times 10^{-13}$ |
| SRG handoff direct product n=2      | $3.12732 \times 10^{-18}$ | $2.00000 \times 10^{-13}$ |
| SRG handoff direct product n=3      | $6.60563 \times 10^{-18}$ | $2.00000 \times 10^{-13}$ |

The SA0 zero-stratum real-Gram output gap is  $1.99334 \times 10^{-2}$ . It is a nonclosure signal, not an approximation error.

**Separation of evidence classes.** Exact rational signs at the six-ring unfolding, integer symmetry dimensions, and source-dependency predicates belong to the retained exact inventory. Finite differences, nonlinear equation residuals, eigenvalue residuals, constructed-metric residuals and quotient comparisons are numerical evidence. The written proofs in the main text do not depend on interpreting one type of measurement as another.

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